

Chapter 4

Vector Spaces and Linear Algebra

This lecture marks our transition from the purely algebraic study of groups to the more geometric realm of linear algebra. Vector spaces are arguably the most tractable algebraic structures—unlike groups, where classification is hopelessly complex, finite-dimensional vector spaces over a fixed field are completely classified by a single integer: the dimension. We begin by establishing the algebraic prerequisites (ring and field homomorphisms, polynomial rings) before developing the theory of vector spaces systematically.

4.1 Ring and Field Homomorphisms

Just as group homomorphisms preserve the group structure, ring and field homomorphisms preserve both operations simultaneously. These maps are fundamental for understanding how algebraic structures relate to one another.

Definition 4.1.1: Ring homomorphism

A **ring homomorphism** is a map $\varphi : R \rightarrow S$ between rings that respects both operations:

1. $\varphi(a + b) = \varphi(a) + \varphi(b)$ for all $a, b \in R$;
2. $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in R$;
3. $\varphi(1_R) = 1_S$.

A **field homomorphism** is a ring homomorphism between fields.

Note:-

Condition (1) implies that φ is a group homomorphism $(R, +) \rightarrow (S, +)$, so we automatically get $\varphi(0) = 0$ and $\varphi(-a) = -\varphi(a)$. However, condition (3) does not follow from condition (2), even for fields! Consider $\varphi = 0$: this satisfies $\varphi(ab) = 0 = 0 \cdot 0 = \varphi(a)\varphi(b)$, but $\varphi(1) = 0 \neq 1$. We must include (3) as a separate axiom.

Proposition 4.1.1 Field homomorphisms are injective

If $\varphi : k \rightarrow K$ is a field homomorphism, then φ is injective.

Proof: Let $a \in k$ with $a \neq 0$. Since k is a field, there exists $b = a^{-1}$ such that $ab = 1_k$. Applying φ :

$$\varphi(a)\varphi(b) = \varphi(ab) = \varphi(1_k) = 1_K \neq 0_K.$$

This implies $\varphi(a) \neq 0_K$ (if $\varphi(a) = 0$, then $\varphi(a)\varphi(b) = 0 \neq 1_K$).

Thus $\ker(\varphi) = \{a \in k : \varphi(a) = 0_K\} = \{0_k\}$, so φ is injective. □

Note:-

This means every field homomorphism is an embedding. There are no “collapsing” maps between fields. This stands in sharp contrast to group and ring homomorphisms, which can have large kernels. The reason is that fields have no nontrivial ideals (a consequence of having multiplicative inverses), so the kernel of any field homomorphism must be either $\{0\}$ or the entire field—but the latter is ruled out by $\varphi(1) = 1 \neq 0$.

4.2 Polynomial Rings and Field Extensions

Polynomials are central objects connecting algebra to analysis, number theory, and geometry. Given a field k , we construct the ring of polynomials $k[x]$ and explore how to enlarge k to contain roots of polynomials.

4.2.1 The polynomial ring $k[x]$

Definition 4.2.1: Polynomial ring

Given a field k , the polynomial ring in one variable is

$$k[x] := \{a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \mid a_i \in k, n \in \mathbb{N}\}.$$

Here x is a **formal variable**—it is not an element of k or any particular set. A polynomial is simply a finite sequence of coefficients, and we add and multiply term by term using the familiar rules.

Note:-

The adjective “formal” is important. When we write $x^2 + 1 \in \mathbb{R}^x$, we are not thinking of x as a real number—we are thinking of $x^2 + 1$ as a symbolic expression. However, we can evaluate a polynomial at any element of k (or any field extension of k): given $f(x) = a_0 + a_1x + \cdots + a_nx^n$ and $r \in k$, we define $f(r) = a_0 + a_1r + \cdots + a_nr^n \in k$.

The polynomial ring $k[x]$ is indeed a ring: it is closed under addition and multiplication, has identity 1, and satisfies all the ring axioms. However, it is not a field.

Proposition 4.2.1 $k[x]$ is not a field

The polynomial ring $k[x]$ has no multiplicative inverses for non-constant polynomials. For instance, x has no inverse in $k[x]$.

Proof: Suppose $f(x) \in k[x]$ satisfies $x \cdot f(x) = 1$. Then $\deg(x \cdot f(x)) = 1 + \deg(f(x)) \geq 1$, but $\deg(1) = 0$. Contradiction. $\square$

4.2.2 The field of rational functions

To make $k[x]$ into a field, we must allow “fractions” of polynomials, just as we pass from $\mathbb{Z}$ to $\mathbb{Q}$.

Definition 4.2.2: Field of rational functions

The field of rational functions over k is

$$k(x) := \left\{ \frac{p}{q} \mid p, q \in k[x], q \neq 0 \right\} / \sim$$

where $\frac{p}{q} \sim \frac{p'}{q'}$ iff $pq' = p'q$. Addition and multiplication are defined as for ordinary fractions.

This construction generalizes naturally to polynomials in any number of variables, giving $k(x_1, \dots, x_n)$, the field of rational functions in n variables.

4.2.3 Formal power series

Another way to extend $k[x]$ is to allow infinite sums—formal power series.

Definition 4.2.3: Formal power series

The ring of formal power series over k is

$$k[[x]] := \left\{ \sum_{i=0}^{\infty} a_i x^i \mid a_i \in k \right\}.$$

Addition and multiplication are defined term by term: given $f = \sum a_i x^i$ and $g = \sum b_i x^i$, we have $f + g = \sum (a_i + b_i) x^i$ and $fg = \sum c_k x^k$ where $c_k = \sum_{i+j=k} a_i b_j$.

Note:-

The product formula makes sense because each coefficient c_k is a finite sum—only finitely many pairs (i, j) satisfy $i + j = k$ with $i, j \geq 0$. This is why we call these “formal” power series: we never worry about convergence. The ring $k[[x]]$ contains $k[x]$ as a subring (polynomials are power series with only finitely many nonzero coefficients).

Unlike $k[x]$, the ring $k[[x]]$ has many more invertible elements.

Lemma 4.2.1 Invertibility in $k[[x]]$

A power series $f = \sum_{i=0}^{\infty} a_i x^i \in k[[x]]$ has a multiplicative inverse if and only if $a_0 \neq 0$.

Proof: $(\Rightarrow)$ If $fg = 1$ for some $g = \sum b_i x^i$, then comparing constant terms gives $a_0 b_0 = 1$, so $a_0 \neq 0$.

$(\Leftarrow)$ Assume $a_0 \neq 0$. We construct $g = \sum b_i x^i$ satisfying $fg = 1$ inductively.

The constant term equation is $a_0 b_0 = 1$, so $b_0 = a_0^{-1}$ (which exists since k is a field).

The coefficient of x^1 equation is $a_0 b_1 + a_1 b_0 = 0$, so $b_1 = -a_0^{-1}(a_1 b_0) = -a_0^{-1} a_1 a_0^{-1}$.

In general, the coefficient of x^n gives $\sum_{i=0}^n a_i b_{n-i} = 0$ for $n \geq 1$, i.e.,

$$a_0 b_n + \sum_{i=1}^n a_i b_{n-i} = 0 \implies b_n = -a_0^{-1} \sum_{i=1}^n a_i b_{n-i}.$$

Each b_n depends only on $b_0, \dots, b_{n-1}$, so we can solve inductively. □

Example 4.2.1

The geometric series: in $k[[x]]$, we have $(1 - x)^{-1} = 1 + x + x^2 + x^3 + \dots = \sum_{i=0}^{\infty} x^i$. This is verified directly: $(1 - x)(1 + x + x^2 + \dots) = 1$ since all higher terms cancel.

4.2.4 Laurent series and field of fractions

Every nonzero element of $k[[x]]$ can be written as $x^m(a_m + a_{m+1}x + \dots)$ where $a_m \neq 0$. To get a field, we need to invert x^m —i.e., allow negative powers.

Definition 4.2.4: Field of Laurent series

The field of Laurent series over k is

$$k((x)) := \left\{ \sum_{i=m}^{\infty} a_i x^i \mid m \in \mathbb{Z}, a_i \in k \right\}.$$

Elements are “power series starting at some (possibly negative) power of x .”

Proposition 4.2.2 $k((x))$ is a field

Every nonzero Laurent series has a multiplicative inverse.

Proof: Let $f = \sum_{i=m}^{\infty} a_i x^i$ with $a_m \neq 0$. Then $f = x^m g$ where $g = a_m + a_{m+1}x + \cdots \in k[[x]]$ has nonzero constant term. By the lemma, g^{-1} exists in $k[[x]]$. Thus $f^{-1} = x^{-m} g^{-1} \in k((x))$. $\square$

4.2.5 Roots of polynomials and field extensions

Given a polynomial $f \in k[x]$ with $\deg(f) \geq 1$, we can ask: does f have a root in k ? If not, can we extend k to a larger field K containing a root?

Definition 4.2.5: Root and field extension

Let $f \in k[x]$ and K be a field containing k . An element $r \in K$ is a root of f if $f(r) = 0$. We call K a field extension of k , written K/k or $k \subset K$.

Example 4.2.2

The polynomial $x^2 - 2 \in \mathbb{Q}^x$ has no roots in $\mathbb{Q}$, but we can form the extension

$$\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\},$$

which is a field containing $\mathbb{Q}$ and the root $\sqrt{2}$. To verify this is a field, we need to check multiplicative inverses: for $a + b\sqrt{2} \neq 0$,

$$\frac{1}{a + b\sqrt{2}} = \frac{a - b\sqrt{2}}{a^2 - 2b^2} = \frac{a}{a^2 - 2b^2} - \frac{b}{a^2 - 2b^2} \sqrt{2} \in \mathbb{Q}(\sqrt{2}).$$

Note that $a^2 - 2b^2 \neq 0$ when $(a, b) \neq (0, 0)$, since $\sqrt{2} \notin \mathbb{Q}$.

Example 4.2.3

Similarly, $x^2 + 1 \in \mathbb{R}^x$ has no real roots. The extension

$$\mathbb{R}(i) = \mathbb{R}(\sqrt{-1}) = \{a + bi : a, b \in \mathbb{R}\} = \mathbb{C}$$

is the field of complex numbers, which contains roots of $x^2 + 1$ (namely $\pm i$).

Definition 4.2.6: Algebraically closed field

A field k is algebraically closed if every non-constant polynomial $f \in k[x]$ has a root in k . Equivalently, every polynomial factors completely into linear factors over k .

Note:-

The Fundamental Theorem of Algebra states that $\mathbb{C}$ is algebraically closed. Over an algebraically closed field, every non-constant polynomial has a root, and there are no further algebraic extensions. This is why $\mathbb{C}$ is often called the “algebraic closure” of $\mathbb{R}$. In contrast, $\mathbb{Q}$ is very far from algebraically closed—there are infinitely many distinct algebraic extensions of $\mathbb{Q}$.

4.3 Characteristic of a Field

Every field k has a distinguished ring homomorphism from the integers, whose kernel determines an important invariant.

Proposition 4.3.1 Universal map from $\mathbb{Z}$

For any field k , there is a unique ring homomorphism $\varphi : \mathbb{Z} \rightarrow k$ defined by $\varphi(1) = 1_k$. Explicitly, $\varphi(n) = n \cdot 1_k$ (add 1_k to itself n times for $n > 0$; negate for $n < 0$).

Proof: Uniqueness: Any ring homomorphism must satisfy $\varphi(1) = 1_k$, and since $\mathbb{Z} = \langle 1 \rangle$ as an additive group, φ is determined by its value on 1.

Existence: Define $\varphi(n) = n \cdot 1_k$. Then $\varphi(m+n) = (m+n) \cdot 1_k = m \cdot 1_k + n \cdot 1_k = \varphi(m) + \varphi(n)$ and $\varphi(mn) = (mn) \cdot 1_k = (m \cdot 1_k)(n \cdot 1_k) = \varphi(m)\varphi(n)$ (verify by induction). Also $\varphi(1) = 1_k$. $\square$

The kernel of φ is a subgroup of $\mathbb{Z}$, hence of the form $n\mathbb{Z}$ for some $n \geq 0$.

Proposition 4.3.2 Kernel is zero or prime

The kernel $\ker(\varphi)$ equals $n\mathbb{Z}$ where either $n = 0$ or $n = p$ is prime.

Proof: Suppose $n > 0$ and n is not prime, so $n = ab$ with $1 < a, b < n$. Then

$$\varphi(n) = \varphi(ab) = \varphi(a)\varphi(b) = 0_k.$$

In a field, $xy = 0$ implies $x = 0$ or $y = 0$. So $\varphi(a) = 0$ or $\varphi(b) = 0$, meaning $a \in \ker(\varphi)$ or $b \in \ker(\varphi)$. But $\ker(\varphi) = n\mathbb{Z}$ and $a, b < n$, so a or b is in $n\mathbb{Z}$ only if $a = 0$ or $b = 0$ —contradicting $a, b > 1$.

Thus if $n > 0$, then n must be prime. $\square$

Definition 4.3.1: Characteristic

The **characteristic** of a field k , denoted $\text{char}(k)$, is:

- 0 if $\ker(\varphi) = \{0\}$, i.e., $\varphi : \mathbb{Z} \hookrightarrow k$ is injective;
- p if $\ker(\varphi) = p\mathbb{Z}$ for a prime p .

Equivalently, $\text{char}(k) = p$ means $\underbrace{1_k + 1_k + \cdots + 1_k}_{p \text{ times}} = 0_k$, and p is the smallest such positive integer.

Example 4.3.1

- $\text{char}(\mathbb{Q}) = \text{char}(\mathbb{R}) = \text{char}(\mathbb{C}) = 0$. The map $\varphi : \mathbb{Z} \rightarrow \mathbb{Q}$ is injective.
- $\text{char}(\mathbb{Z}/p) = p$ for prime p . Indeed, $p \cdot 1 = 0$ in $\mathbb{Z}/p$.
- $\text{char}(k(x)) = \text{char}(k[[x]]) = \text{char}(k((x))) = \text{char}(k)$.

Note:-

The field $\mathbb{Z}/p$ is often denoted $\mathbb{F}_p$ and called the **prime field** of characteristic p . Every field of characteristic p contains a copy of $\mathbb{F}_p$; every field of characteristic 0 contains a copy of $\mathbb{Q}$. Thus $\mathbb{F}_p$ and $\mathbb{Q}$ are the “minimal” fields in their respective characteristics.

Theorem 4.3.1 Existence and uniqueness of finite fields

For every prime p and $n \geq 1$, there exists a unique (up to isomorphism) field with exactly p^n elements, denoted $\mathbb{F}_{p^n}$ or $\text{GF}(p^n)$. These are all the finite fields.

Note:-

We will not prove this theorem here. The finite field $\mathbb{F}_{p^n}$ can be constructed as a splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$. There are also infinite fields of characteristic p —for example, $\mathbb{F}_p(x)$, the field of rational functions over $\mathbb{F}_p$.

4.4 Vector Spaces

We now arrive at the central structure of linear algebra. A vector space is a set with two operations—addition and scalar multiplication—satisfying axioms that generalize familiar properties of $\mathbb{R}^n$.

Definition 4.4.1: Vector space

Fix a field k . A **vector space over k** (or **k -vector space**) is a set V equipped with two operations:

1. **Vector addition:** $+: V \times V \rightarrow V$;
2. **Scalar multiplication:** $\cdot: k \times V \rightarrow V$;

satisfying the following axioms:

1. $(V, +)$ is an abelian group with identity 0 (the **zero vector**);
2. $1 \cdot v = v$ for all $v \in V$ (identity for scalar multiplication);
3. $(ab) \cdot v = a \cdot (b \cdot v)$ for all $a, b \in k$ and $v \in V$ (associativity);
4. $(a + b) \cdot v = a \cdot v + b \cdot v$ for all $a, b \in k$ and $v \in V$ (distributivity over scalar addition);
5. $a \cdot (v + w) = a \cdot v + a \cdot w$ for all $a \in k$ and $v, w \in V$ (distributivity over vector addition).

Elements of V are called **vectors**; elements of k are called **scalars**.

Note:-

The distributive laws (4) and (5) connect the two operations. Note that axiom (1) implies $0_V + v = v$ and $v + (-v) = 0_V$, while axiom (2)-(5) give us things like $0_k \cdot v = 0_V$ (proof: $0_k \cdot v = (0_k + 0_k) \cdot v = 0_k \cdot v + 0_k \cdot v$, so $0_k \cdot v = 0_V$ by cancellation). We also get $(-1) \cdot v = -v$.

4.4.1 Fundamental examples

Example 4.4.1

The standard vector space k^n : For any $n \geq 1$, the set

$$k^n = \{(a_1, a_2, \dots, a_n) : a_i \in k\}$$

is a vector space with componentwise operations:

$$(a_1, \dots, a_n) + (b_1, \dots, b_n) = (a_1 + b_1, \dots, a_n + b_n),$$

$$c \cdot (a_1, \dots, a_n) = (ca_1, \dots, ca_n).$$

This is the prototypical n -dimensional vector space.

Example 4.4.2

Infinite sequences k^∞ : The set of all sequences in k ,

$$k^\infty = \{(a_i)_{i \in \mathbb{N}} : a_i \in k\} = \{(a_1, a_2, a_3, \dots) : a_i \in k\},$$

is a vector space with componentwise operations. It contains the subspace of eventually zero sequences—those with $a_i = 0$ for all sufficiently large i .

Example 4.4.3

Function spaces: For any set S , the set

$$k^S = \{f : S \rightarrow k\}$$

of all functions from S to k is a vector space with pointwise operations: $(f + g)(s) = f(s) + g(s)$ and $(c \cdot f)(s) = c \cdot f(s)$.

When $S = \mathbb{N}$, we recover k^∞ . When $S = \{1, \dots, n\}$, we recover k^n .

Example 4.4.4

Polynomial and power series spaces:

- $k[x]$ is a k -vector space (addition is polynomial addition; scalar multiplication is multiplying each coefficient by the scalar).
- $k[[x]]$ is a k -vector space.
- For each $d \geq 0$, the set $k[x]_{\leq d} = \{f \in k[x] : \deg(f) \leq d\}$ of polynomials of degree at most d is a k -vector space.

Example 4.4.5

Spaces of maps $\mathbb{R} \rightarrow \mathbb{R}$: Consider $V = \{f : \mathbb{R} \rightarrow \mathbb{R}\}$, all functions from $\mathbb{R}$ to $\mathbb{R}$. This contains many interesting subspaces:

- Continuous functions $C(\mathbb{R})$;
- Differentiable functions $C^1(\mathbb{R})$;
- Smooth (infinitely differentiable) functions $C^\infty(\mathbb{R})$;
- Polynomials $\mathbb{R}^x$.

Each is closed under addition and scalar multiplication, forming a vector space.

4.4.2 Subspaces

Definition 4.4.2: Subspace

A **subspace** of a vector space V is a nonempty subset $W \subseteq V$ that is closed under addition and scalar multiplication:

1. $w_1, w_2 \in W \implies w_1 + w_2 \in W$;
2. $w \in W, a \in k \implies a \cdot w \in W$.

Equivalently, W is a subspace if it is a vector space under the inherited operations.

Proposition 4.4.1 Subspace criterion

A nonempty subset $W \subseteq V$ is a subspace if and only if $aw_1 + bw_2 \in W$ for all $a, b \in k$ and $w_1, w_2 \in W$.

Proof: ($\Rightarrow$) If W is a subspace, then $aw_1 \in W$ and $bw_2 \in W$ by scalar multiplication closure, and $aw_1 + bw_2 \in W$ by addition closure.

($\Leftarrow$) Taking $a = b = 1$ gives closure under addition. Taking $b = 0$ gives closure under scalar multiplication. $\square$

Note:-

The condition “nonempty” is required. Alternatively, we can replace it with “ $0 \in W$,” which follows from taking $a = 0$ in the scalar multiplication closure. So a subspace is precisely a nonempty subset closed under linear combinations.

Example 4.4.6

- $\{0\}$ and V itself are always subspaces (the **trivial subspaces**).
- In $\mathbb{R}^3$, subspaces are: $\{0\}$, lines through the origin, planes through the origin, and $\mathbb{R}^3$.
- In $k[x]$, the polynomials of degree $\leq n$ form a subspace. The polynomials of degree exactly n do not form a subspace (not closed under addition: $x^n + (-x^n) = 0$).

4.4.3 Direct Sums of Subspaces

When can we decompose a vector space into “independent” pieces? The notion of direct sum captures exactly this.

Definition 4.4.3: Sum of subspaces

Given subspaces $U, W \subseteq V$, their **sum** is

$$U + W = \{u + w : u \in U, w \in W\}.$$

This is the smallest subspace of V containing both U and W .

Proposition 4.4.2 Sum is a subspace

For subspaces $U, W \subseteq V$, the sum $U + W$ is a subspace of V .

Proof: Let $v_1 = u_1 + w_1$ and $v_2 = u_2 + w_2$ be elements of $U + W$, and let $\lambda \in k$. Then

$$v_1 + v_2 = (u_1 + u_2) + (w_1 + w_2) \in U + W$$

since $u_1 + u_2 \in U$ and $w_1 + w_2 \in W$. Similarly, $\lambda v_1 = \lambda u_1 + \lambda w_1 \in U + W$. The zero vector $0 = 0 + 0 \in U + W$. $\square$

Definition 4.4.4: Direct sum of subspaces

The sum $U + W$ is called a **direct sum**, written $U \oplus W$, if $U \cap W = \{0\}$. Equivalently, every element of $U + W$ can be written uniquely as $u + w$ with $u \in U$ and $w \in W$.

Proposition 4.4.3 Characterizations of direct sum

For subspaces $U, W \subseteq V$, the following are equivalent:

1. $U + W$ is a direct sum (i.e., $U \cap W = \{0\}$).
2. Every $v \in U + W$ has a unique representation $v = u + w$ with $u \in U$, $w \in W$.
3. $u + w = 0$ with $u \in U$, $w \in W$ implies $u = w = 0$.

Proof: (1) $\Rightarrow$ (2): Suppose $v = u_1 + w_1 = u_2 + w_2$. Then $u_1 - u_2 = w_2 - w_1$. The left side is in U , the right side is in W , so both are in $U \cap W = \{0\}$. Thus $u_1 = u_2$ and $w_1 = w_2$.

(2) $\Rightarrow$ (3): If $u + w = 0$, then $u + w = 0 + 0$. By uniqueness, $u = 0$ and $w = 0$.

(3) $\Rightarrow$ (1): If $v \in U \cap W$, then $v + (-v) = 0$ with $v \in U$ and $-v \in W$. By (3), $v = 0$. $\square$

Theorem 4.4.1 Dimension formula for sums

For finite-dimensional subspaces $U, W \subseteq V$:

$$\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W).$$

In particular, if $U + W = U \oplus W$ is a direct sum, then $\dim(U \oplus W) = \dim(U) + \dim(W)$.

Proof: Let $\{v_1, \dots, v_k\}$ be a basis of $U \cap W$. Extend to a basis $\{v_1, \dots, v_k, u_1, \dots, u_m\}$ of U and to a basis $\{v_1, \dots, v_k, w_1, \dots, w_n\}$ of W .

Claim: $\{v_1, \dots, v_k, u_1, \dots, u_m, w_1, \dots, w_n\}$ is a basis of $U + W$.

Spanning: Any $x \in U + W$ is $x = u + w$ with $u \in U$, $w \in W$. Express u and w in terms of the bases above; the result is a linear combination of all the listed vectors.

Independence: Suppose $\sum a_i v_i + \sum b_j u_j + \sum c_\ell w_\ell = 0$. Then

$$\sum c_\ell w_\ell = -\sum a_i v_i - \sum b_j u_j \in U.$$

But $\sum c_\ell w_\ell \in W$, so $\sum c_\ell w_\ell \in U \cap W = \text{Span}\{v_1, \dots, v_k\}$. Since $\{v_1, \dots, v_k, w_1, \dots, w_n\}$ is independent, all $c_\ell = 0$. Similarly, all $b_j = 0$, and then all $a_i = 0$.

Thus $\dim(U + W) = k + m + n = (k + m) + (k + n) - k = \dim(U) + \dim(W) - \dim(U \cap W)$. $\square$

Definition 4.4.5: Internal direct sum decomposition

We say V is the **(internal) direct sum** of subspaces U and W , written $V = U \oplus W$, if:

1. $V = U + W$ (every vector is a sum of elements from U and W), and
2. $U \cap W = \{0\}$ (the decomposition is unique).

More generally, $V = U_1 \oplus U_2 \oplus \dots \oplus U_n$ if every $v \in V$ can be written uniquely as $v = u_1 + u_2 + \dots + u_n$ with $u_i \in U_i$.

Example 4.4.7

- In $\mathbb{R}^3$, let $U = \{(x, 0, 0) : x \in \mathbb{R}\}$ (the x -axis) and $W = \{(0, y, z) : y, z \in \mathbb{R}\}$ (the yz -plane). Then $\mathbb{R}^3 = U \oplus W$.
- Let $V = k^n$ and $U = \text{Span}\{e_1, \dots, e_m\}$, $W = \text{Span}\{e_{m+1}, \dots, e_n\}$. Then $k^n = U \oplus W$.
- If V is the space of $n \times n$ matrices, let S = symmetric matrices and A = antisymmetric matrices. Then $V = S \oplus A$ (any matrix $M = \frac{1}{2}(M + M^T) + \frac{1}{2}(M - M^T)$).

Note:-

When we diagonalize a linear operator, we decompose the vector space as a direct sum of eigenspaces. When we study quotient spaces, we often use the fact that $V = U \oplus W$ implies $V/U \cong W$; if that doesn't make sense right now, don't worry, we'll get to it. The direct sum structure makes computations tractable by breaking a problem into independent pieces.

4.4.4 External Direct Sums and Products

We can also construct new vector spaces from old ones by taking products or direct sums “externally.”

Definition 4.4.6: Direct product (external direct sum)

Given vector spaces $V_1, \dots, V_n$ over the same field k , their **direct product** (or **external direct sum**) is

$$V_1 \times V_2 \times \cdots \times V_n = \{(v_1, v_2, \dots, v_n) : v_i \in V_i\}$$

with componentwise operations:

$$\begin{aligned} (v_1, \dots, v_n) + (w_1, \dots, w_n) &= (v_1 + w_1, \dots, v_n + w_n), \\ \lambda(v_1, \dots, v_n) &= (\lambda v_1, \dots, \lambda v_n). \end{aligned}$$

This is also denoted $V_1 \oplus V_2 \oplus \cdots \oplus V_n$ or $\bigoplus_{i=1}^n V_i$.

Proposition 4.4.4 Dimension of direct product

If each V_i is finite-dimensional, then

$$\dim(V_1 \times \cdots \times V_n) = \dim(V_1) + \cdots + \dim(V_n).$$

Proof: If $\mathcal{B}_i = \{v_{i,1}, \dots, v_{i,d_i}\}$ is a basis of V_i , then

$$\{(v_{1,1}, 0, \dots, 0), \dots, (v_{1,d_1}, 0, \dots, 0), (0, v_{2,1}, 0, \dots, 0), \dots, (0, \dots, 0, v_{n,d_n})\}$$

is a basis of $V_1 \times \cdots \times V_n$. This has $d_1 + \cdots + d_n$ elements. □

Note:-

The external direct sum $V_1 \times V_2$ and the internal direct sum $U \oplus W \subseteq V$ are related as follows: if $V = U \oplus W$, then the map

$$U \times W \rightarrow V, \quad (u, w) \mapsto u + w$$

is an isomorphism. Conversely, $V_1 \times V_2$ contains copies of V_1 and V_2 as subspaces (embedded as $V_1 \times \{0\}$ and $\{0\} \times V_2$), and $V_1 \times V_2$ is their internal direct sum.

Example 4.4.8

The space $\mathbb{R}^{m+n}$ is isomorphic to $\mathbb{R}^m \times \mathbb{R}^n$ via $(x_1, \dots, x_m, y_1, \dots, y_n) \mapsto ((x_1, \dots, x_m), (y_1, \dots, y_n))$. This is why we freely identify $\mathbb{R}^m \oplus \mathbb{R}^n$ with $\mathbb{R}^{m+n}$.

4.5 Span, Linear Independence, and Basis

These three concepts—span, linear independence, and basis—are the building blocks of linear algebra. They allow us to describe vector spaces in terms of generating sets and coordinates.

4.5.1 Linear combinations and span

Definition 4.5.1: Linear combination

A **linear combination** of vectors $v_1, \dots, v_n \in V$ is any vector of the form

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n$$

where $a_1, \dots, a_n \in k$ are scalars.

Definition 4.5.2: Span

Given vectors $v_1, \dots, v_n \in V$, the **span** of $\{v_1, \dots, v_n\}$ is the set of all linear combinations:

$$\text{Span}(v_1, \dots, v_n) = \{a_1 v_1 + \dots + a_n v_n : a_i \in k\}.$$

This is the **smallest subspace** of V containing $v_1, \dots, v_n$.

We say $v_1, \dots, v_n$ **span** V (or are a **spanning set** for V) if $\text{Span}(v_1, \dots, v_n) = V$.

Proposition 4.5.1 Span is a subspace

For any $v_1, \dots, v_n \in V$, the span $\text{Span}(v_1, \dots, v_n)$ is a subspace of V .

Proof: Let $W = \text{Span}(v_1, \dots, v_n)$. Clearly $0 = 0 \cdot v_1 + \dots + 0 \cdot v_n \in W$, so $W \neq \emptyset$. For closure: if $w = \sum a_i v_i$ and $w' = \sum a'_i v_i$ are in W , and $c, c' \in k$, then

$$cw + c'w' = \sum (ca_i + c'a'_i)v_i \in W.$$

□

The span can also be defined for infinite (or even arbitrary) sets of vectors.

Definition 4.5.3: Span of an arbitrary set

For $S \subseteq V$, the **span** of S is the set of all **finite** linear combinations of elements of S :

$$\text{Span}(S) = \left\{ \sum_{i=1}^n a_i v_i : n \geq 0, a_i \in k, v_i \in S \right\}.$$

When $n = 0$, the empty sum is 0, so $0 \in \text{Span}(S)$ always.

4.5.2 Linear independence

Definition 4.5.4: Linear independence

Vectors $v_1, \dots, v_n \in V$ are **linearly independent** if the only way to write 0 as a linear combination is the trivial way:

$$a_1v_1 + a_2v_2 + \dots + a_nv_n = 0 \implies a_1 = a_2 = \dots = a_n = 0.$$

Otherwise, they are **linearly dependent**—there exist scalars, not all zero, such that $\sum a_i v_i = 0$.

Note:-

Linear dependence means some vector can be expressed as a linear combination of the others. If $a_1v_1 + \dots + a_nv_n = 0$ with (say) $a_1 \neq 0$, then

$$v_1 = -\frac{a_2}{a_1}v_2 - \dots - \frac{a_n}{a_1}v_n.$$

So v_1 is “redundant”—it’s already in the span of the others.

Proposition 4.5.2 Equivalent characterizations of linear independence

The following are equivalent for $v_1, \dots, v_n \in V$:

1. $v_1, \dots, v_n$ are linearly independent.
2. No v_i is a linear combination of the others.
3. The linear map $\phi : k^n \rightarrow V$ defined by $\phi(a_1, \dots, a_n) = \sum a_i v_i$ is injective.

Proof: (1) $\Leftrightarrow$ (3): $\ker(\phi) = \{(a_1, \dots, a_n) : \sum a_i v_i = 0\}$. Linear independence says $\ker(\phi) = \{0\}$, which is equivalent to ϕ being injective.

(1) $\Rightarrow$ (2): Suppose $v_j = \sum_{i \neq j} b_i v_i$. Then $\sum_{i \neq j} b_i v_i + (-1)v_j = 0$ is a nontrivial linear combination (coefficient of v_j is $-1 \neq 0$), contradicting independence.

(2) $\Rightarrow$ (1): Suppose $\sum a_i v_i = 0$ with some $a_j \neq 0$. Then $v_j = -a_j^{-1} \sum_{i \neq j} a_i v_i$, contradicting (2). $\square$

Similarly, we can characterize spanning:

Proposition 4.5.3 Characterization of spanning

The following are equivalent:

1. $v_1, \dots, v_n$ span V .
2. Every $v \in V$ can be written as $\sum a_i v_i$ for some $a_i \in k$.
3. The linear map $\phi : k^n \rightarrow V$ defined by $\phi(a_1, \dots, a_n) = \sum a_i v_i$ is surjective.

4.5.3 Basis

Definition 4.5.5: Basis

A **basis** for V is a set $\{v_1, \dots, v_n\}$ of vectors that are both linearly independent and span V . Equivalently, the linear map $\phi : k^n \rightarrow V$ is **bijective**.

Proposition 4.5.4 Unique representation

If $\{v_1, \dots, v_n\}$ is a basis for V , then every vector $v \in V$ can be written uniquely as

$$v = a_1 v_1 + a_2 v_2 + \cdots + a_n v_n$$

for some $a_i \in k$. The scalars $(a_1, \dots, a_n)$ are called the coordinates of v with respect to this basis.

Proof: Existence: Since $\{v_i\}$ spans V , every v is some linear combination.

Uniqueness: If $v = \sum a_i v_i = \sum a'_i v_i$, then $\sum (a_i - a'_i) v_i = 0$. By linear independence, $a_i - a'_i = 0$ for all i , so $a_i = a'_i$. $\square$

Example 4.5.1

Standard basis of k^n : The vectors

$$e_1 = (1, 0, \dots, 0), \quad e_2 = (0, 1, 0, \dots, 0), \quad \dots, \quad e_n = (0, \dots, 0, 1)$$

form the standard basis of k^n . Any $(a_1, \dots, a_n) = a_1 e_1 + \cdots + a_n e_n$.

Example 4.5.2

Non-standard basis of k^2 : In k^2 , the vectors $(1, 0)$ and $(0, 1)$ form the standard basis. But so do $(1, 1)$ and $(1, -1)$ (for most k —we need $\text{char}(k) \neq 2$ for these to be independent).

Example 4.5.3

Basis of $k[x]_{\leq n}$: The polynomials $\{1, x, x^2, \dots, x^n\}$ form a basis for $k[x]_{\leq n}$, the space of polynomials of degree $\leq n$.

Question 1

1. Does $k[x]$ have a finite basis? What is a basis for $k[x]$?
2. Does $k[[x]]$ have a basis?

Solution

(1) The polynomial ring $k[x]$ does not have a finite basis, because any finite set $\{p_1, \dots, p_n\}$ of polynomials has bounded degree, so cannot span polynomials of higher degree.

However, $k[x]$ has the infinite basis $\{1, x, x^2, x^3, \dots\}$. Every polynomial is a finite linear combination of these monomials, and they are linearly independent (a nontrivial linear combination would give a nonzero polynomial equaling 0).

(2) This is subtle. By the axiom of choice, $k[[x]]$ has a basis (every vector space does). But this basis cannot be explicitly described—it is uncountably infinite and highly non-constructive. Unlike $k[x]$, the monomials $\{1, x, x^2, \dots\}$ do not form a basis of $k[[x]]$: while they are linearly independent, they do not span (a power series like $\sum_{i=0}^{\infty} x^i$ is not a finite linear combination of monomials).

4.6 Finite-Dimensional Vector Spaces and Dimension

The notion of dimension will appear constantly throughout linear algebra and beyond. Unlike groups, where classification is intractable, finite-dimensional vector spaces are completely determined by a single integer—their dimension. We establish this remarkable fact through a sequence of lemmas about the interplay between spanning sets and linearly independent sets.

Definition 4.6.1: Finite-dimensional

A vector space V is **finite-dimensional** if there exists a finite spanning set for V , i.e., vectors $v_1, \dots, v_n$ such that $\text{Span}(v_1, \dots, v_n) = V$. Otherwise, V is **infinite-dimensional**.

4.6.1 Extracting bases from spanning sets

Our first key result shows that every spanning set contains a basis.

Lemma 4.6.1 Spanning sets contain bases

If $\{v_1, \dots, v_n\}$ spans V , then some subset of $\{v_1, \dots, v_n\}$ is a basis for V .

Proof: We construct a basis by iteratively removing redundant vectors.

If $\{v_1, \dots, v_n\}$ is linearly independent, we are done—it is already a basis.

Otherwise, there exists a nontrivial relation $\sum_{i=1}^n a_i v_i = 0$ with some $a_j \neq 0$. Then

$$v_j = -\frac{1}{a_j} \sum_{i \neq j} a_i v_i \in \text{Span}(\{v_i : i \neq j\}).$$

So $\{v_i : i \neq j\}$ still spans V (any linear combination involving v_j can be rewritten using the other vectors).

Remove v_j and repeat. Since we start with finitely many vectors, this process terminates. When it does, the remaining vectors are linearly independent (else we could remove another) and still span V , hence form a basis. $\square$

Corollary 4.6.1 Finite-dimensional spaces have bases

Every finite-dimensional vector space has a basis.

4.6.2 Extending linearly independent sets to bases

The dual result shows that any linearly independent set can be enlarged to a basis.

Lemma 4.6.2 Linearly independent sets extend to bases

If V is finite-dimensional and $\{v_1, \dots, v_m\}$ is linearly independent, then there exists a basis of V containing $\{v_1, \dots, v_m\}$.

Proof: Let $\{w_1, \dots, w_r\}$ be a finite spanning set for V (exists by finite-dimensionality). We enlarge $\{v_1, \dots, v_m\}$ to a basis using vectors from $\{w_j\}$.

Define $W_0 = \text{Span}(v_1, \dots, v_m)$, with basis $\{v_1, \dots, v_m\}$ (our independent set).

Induction step: Suppose we have constructed a basis $\mathcal{B}_{j-1}$ for $W_{j-1} = \text{Span}(v_1, \dots, v_m, w_{i_1}, \dots, w_{i_{j-1}})$.

Consider w_j :

- If $w_j \in W_{j-1}$, set $W_j = W_{j-1}$ and $\mathcal{B}_j = \mathcal{B}_{j-1}$.
- If $w_j \notin W_{j-1}$, then $\mathcal{B}_{j-1} \cup \{w_j\}$ is linearly independent (otherwise w_j would be a linear combination of $\mathcal{B}_{j-1}$, contradicting $w_j \notin W_{j-1}$). Set $W_j = \text{Span}(\mathcal{B}_{j-1} \cup \{w_j\})$ and $\mathcal{B}_j = \mathcal{B}_{j-1} \cup \{w_j\}$.

After processing all w_j , we have $W_r = \text{Span}(v_1, \dots, v_m, w_1, \dots, w_r) \supseteq \text{Span}(w_1, \dots, w_r) = V$. So $\mathcal{B}_r$ is a linearly independent set spanning V , i.e., a basis containing $\{v_1, \dots, v_m\}$. $\square$

4.6.3 Invariance of basis size

The following theorem shows that the size of a basis is intrinsic to the vector space, not dependent on the choice of basis.

Theorem 4.6.1 All bases have the same size

If V has a basis with n elements and another basis with m elements, then $m = n$.

Proof: Let $\{v_1, \dots, v_n\}$ and $\{w_1, \dots, w_m\}$ be bases for V . We show $m \leq n$ and $n \leq m$, hence $m = n$.

Claim: $\{v_1, \dots, v_{m-1}, w_j\}$ is a basis for some choice of reordering, for each j we process.

We proceed by a “replacement” argument. Start with the basis $\{v_1, \dots, v_n\}$.

Since $w_1 \in V = \text{Span}(v_1, \dots, v_n)$, we can write $w_1 = \sum_{i=1}^n a_i v_i$. Not all $a_i = 0$ (else $w_1 = 0$, contradicting independence of $\{w_j\}$). By reordering, assume $a_n \neq 0$.

Then

$$v_n = a_n^{-1}(w_1 - \sum_{i=1}^{n-1} a_i v_i) \in \text{Span}(v_1, \dots, v_{n-1}, w_1).$$

So

$$\text{Span}(v_1, \dots, v_{n-1}, w_1) \supseteq \text{Span}(v_1, \dots, v_n) = V,$$

and thus $\{v_1, \dots, v_{n-1}, w_1\}$ spans V . Since it has n elements and spans, it contains a basis—but removing any element would give $n - 1$ vectors. We verify it’s actually linearly independent: if

$$\sum_{i=1}^{n-1} b_i v_i + c w_1 = 0$$

with $c \neq 0$, then $w_1 \in \text{Span}(v_1, \dots, v_{n-1})$, so $v_n \in \text{Span}(v_1, \dots, v_{n-1})$, contradicting independence of $\{v_i\}$. Thus $c = 0$, and then all $b_i = 0$. So $\{v_1, \dots, v_{n-1}, w_1\}$ is a basis.

Repeat: $w_2 = \sum_{i=1}^{n-1} a_i v_i + c w_1$. Some $a_i \neq 0$ (else $w_2 \in \text{Span}(w_1)$, contradicting independence). Replace that v_i with w_2 .

Continue this process, replacing one v with one w each time. After m steps, we have a basis

$$\{v_{i_1}, \dots, v_{i_{n-m}}, w_1, \dots, w_m\}$$

(assuming $m \leq n$).

If $m > n$, the process would terminate before incorporating all w_j , but then some w_j would be in the span of earlier w ’s, contradicting independence. Hence $m \leq n$.

By symmetry, $n \leq m$. Thus $m = n$. □

Definition 4.6.2: Dimension

The **dimension** of a finite-dimensional vector space V , denoted $\dim(V)$ or $\dim_k(V)$, is the number of elements in any basis. The theorem above shows this is well-defined.

By convention, $\dim(\{0\}) = 0$ (the empty set is a basis for the zero space).

Corollary 4.6.2 Isomorphism with k^n

Every finite-dimensional vector space V with $\dim(V) = n$ is isomorphic to k^n .

Proof: Let $\{v_1, \dots, v_n\}$ be a basis for V . The map

$$\phi : k^n \rightarrow V, \quad (a_1, \dots, a_n) \mapsto \sum_{i=1}^n a_i v_i$$

is linear (easily verified), injective (by linear independence), and surjective (by spanning). Hence ϕ is an isomorphism. $\square$

Note:-

The isomorphism $\phi : k^n \xrightarrow{\sim} V$ depends on the choice of basis. Different bases give different isomorphisms. This is the key insight of linear algebra: choosing a basis amounts to choosing coordinates, and we represent elements of V by column vectors $\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in k^n$.

Example 4.6.1

Dimensions:

- $\dim(k^n) = n$. The standard basis has n elements.
- $\dim(k[x]_{\leq d}) = d + 1$. The monomials $\{1, x, \dots, x^d\}$ form a basis.
- $k[x]$ is infinite-dimensional: any finite set of polynomials has bounded degree and cannot span.
- $k[[x]]$ is infinite-dimensional (and in a “larger” sense—its bases are uncountable).

4.7 Linear Maps

Having established the structure of vector spaces, we now study the morphisms between them. Just as group homomorphisms preserve the group operation, linear maps preserve both vector space operations.

Definition 4.7.1: Linear map

Let V and W be vector spaces over k . A **linear map** (or k -linear map, or homomorphism of vector spaces) is a function $\varphi : V \rightarrow W$ satisfying:

1. $\varphi(u + v) = \varphi(u) + \varphi(v)$ for all $u, v \in V$ (additivity);
2. $\varphi(\lambda v) = \lambda \varphi(v)$ for all $\lambda \in k$ and $v \in V$ (homogeneity).

Equivalently, φ preserves all linear combinations:

$$\varphi \left(\sum_{i=1}^n a_i v_i \right) = \sum_{i=1}^n a_i \varphi(v_i).$$

Note:-

A linear map is completely determined by its values on a basis. If $\{v_1, \dots, v_n\}$ is a basis for V , and we specify $\varphi(v_i) = w_i \in W$ for each i , then for any $v = \sum a_i v_i$:

$$\varphi(v) = \varphi \left(\sum a_i v_i \right) = \sum a_i \varphi(v_i) = \sum a_i w_i.$$

This is the **universal property** of bases: specifying a linear map from V is equivalent to choosing where to send the basis vectors.

Definition 4.7.2: Space of linear maps

The set of all linear maps from V to W is denoted

$$\text{Hom}_k(V, W) \quad \text{or simply} \quad \text{Hom}(V, W).$$

When $V = W$, we write $\text{End}(V) = \text{Hom}(V, V)$ and call elements endomorphisms.

Proposition 4.7.1 $\text{Hom}(V, W)$ is a vector space

The set $\text{Hom}(V, W)$ is itself a vector space over k , with operations:

- **Addition:** $(\varphi + \psi)(v) = \varphi(v) + \psi(v)$ for all $v \in V$;
- **Scalar multiplication:** $(\lambda\varphi)(v) = \lambda \cdot \varphi(v)$ for all $v \in V$.

Proof: We must verify that $\varphi + \psi$ and $\lambda\varphi$ are linear maps when φ, ψ are.

For $\varphi + \psi$: Let $u, v \in V$ and $\mu \in k$.

$$\begin{aligned} (\varphi + \psi)(u + v) &= \varphi(u + v) + \psi(u + v) = \varphi(u) + \varphi(v) + \psi(u) + \psi(v) \\ &= (\varphi(u) + \psi(u)) + (\varphi(v) + \psi(v)) = (\varphi + \psi)(u) + (\varphi + \psi)(v). \\ (\varphi + \psi)(\mu v) &= \varphi(\mu v) + \psi(\mu v) = \mu\varphi(v) + \mu\psi(v) = \mu(\varphi(v) + \psi(v)) = \mu(\varphi + \psi)(v). \end{aligned}$$

For $\lambda\varphi$: Similarly,

$$\begin{aligned} (\lambda\varphi)(u + v) &= \lambda(\varphi(u + v)) = \lambda(\varphi(u) + \varphi(v)) = \lambda\varphi(u) + \lambda\varphi(v) = (\lambda\varphi)(u) + (\lambda\varphi)(v). \\ (\lambda\varphi)(\mu v) &= \lambda(\varphi(\mu v)) = \lambda(\mu\varphi(v)) = \mu(\lambda\varphi(v)) = \mu(\lambda\varphi)(v). \end{aligned}$$

The vector space axioms for $\text{Hom}(V, W)$ follow from those of W (operations are defined pointwise). The zero vector is the zero map $v \mapsto 0_W$, and additive inverses are $(-\varphi)(v) = -\varphi(v)$. $\square$

Note:-

The verification that $\varphi + \psi$ and $\lambda\varphi$ are linear is “rather boring, but worth checking if you’re not sure.” The operations on $\text{Hom}(V, W)$ are inherited pointwise from W , so the axioms follow routinely.

Theorem 4.7.1 Dimension of $\text{Hom}(V, W)$

If $\dim(V) = n$ and $\dim(W) = m$, then

$$\dim(\text{Hom}(V, W)) = mn.$$

The proof of this theorem will emerge naturally when we establish the correspondence between linear maps and matrices in the next section.

Note:-

We will soon see: if $\dim(V) = n$ and $\dim(W) = m$, then $\dim(\text{Hom}(V, W)) = mn$. In terms of bases for V and W , linear maps become $m \times n$ matrices!

4.7.1 Dependence on the scalar field

The notions of basis, dimension, and linearity depend crucially on the choice of scalar field k . The same set can be a vector space over different fields, with different dimensions.

Example 4.7.1

$\mathbb{C}$ as a vector space over different fields: Consider $\mathbb{C}$ as a vector space:

- Over $\mathbb{C}$: $\dim_{\mathbb{C}}(\mathbb{C}) = 1$, with basis $\{1\}$. Every complex number z is $z \cdot 1$.
- Over $\mathbb{R}$: $\dim_{\mathbb{R}}(\mathbb{C}) = 2$, with basis $\{1, i\}$. Every complex number $a + bi$ is the $\mathbb{R}$ -linear combination $a \cdot 1 + b \cdot i$.

Note:-

More generally, if $k' \subset k$ is a subfield (e.g., $\mathbb{R} \subset \mathbb{C}$ or $\mathbb{Q} \subset \mathbb{R}$), then any k -vector space V is automatically a k' -vector space by “restriction of scalars”—we simply forget that we can multiply by elements of $k \setminus k'$. The k' -dimension is always at least as large as the k -dimension.

Proposition 4.7.2 Linearity over subfields

If V and W are k -vector spaces and $k' \subset k$ is a subfield, then:

1. V is a k' -vector space (by restriction of scalars).
2. Any k -linear map $\varphi : V \rightarrow W$ is also k' -linear.
3. The converse fails: not every k' -linear map is k -linear.

Thus $\text{Hom}_k(V, W) \subsetneq \text{Hom}_{k'}(V, W)$ in general.

Example 4.7.2

Complex conjugation is $\mathbb{R}$ -linear but not $\mathbb{C}$ -linear: Consider the map

$$\bar{} : \mathbb{C} \rightarrow \mathbb{C}, \quad z = a + bi \mapsto \bar{z} = a - bi.$$

$\mathbb{R}$ -linearity: For $z_1, z_2 \in \mathbb{C}$ and $r \in \mathbb{R}$:

- $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ ✓
- $\overline{r \cdot z} = r \cdot \bar{z}$ (since $\bar{r} = r$ for $r \in \mathbb{R}$) ✓

Not $\mathbb{C}$ -linear: For $\lambda = i$ and $z = 1$:

$$\overline{i \cdot 1} = \bar{i} = -i, \quad \text{but} \quad i \cdot \bar{1} = i \cdot 1 = i.$$

Since $-i \neq i$, complex conjugation is not $\mathbb{C}$ -linear.

This example illustrates a profound point: the “natural” maps between spaces can depend on what scalars we consider. Complex conjugation is an $\mathbb{R}$ -linear automorphism of $\mathbb{C}$ that is not $\mathbb{C}$ -linear—it does not commute with multiplication by i .

4.8 Matrix Representation of Linear Maps

The correspondence between linear maps and matrices is one of the most powerful ideas in linear algebra. It reduces abstract questions about linear maps to concrete computations with arrays of numbers.

4.8.1 The matrix of a linear map

Fix bases $\mathcal{B} = (v_1, \dots, v_n)$ for V and $\mathcal{C} = (w_1, \dots, w_m)$ for W . Given a linear map $\varphi : V \rightarrow W$, we can express each image $\varphi(v_j)$ as a linear combination of the basis vectors of W :

$$\varphi(v_j) = \sum_{i=1}^m a_{ij} w_i = a_{1j} w_1 + a_{2j} w_2 + \cdots + a_{mj} w_m.$$

Definition 4.8.1: Matrix of a linear map

The **matrix of φ** with respect to bases $\mathcal{B}$ and $\mathcal{C}$ is the $m \times n$ matrix

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \in \mathcal{M}_{m \times n}(k).$$

The j -th **column** of A is the coordinate vector of $\varphi(v_j)$ in the basis $\mathcal{C}$.

Note:-

The key insight: **columns of A are the images of basis vectors, expressed in coordinates.** This is worth memorizing: the j -th column of the matrix tells you where the j -th basis vector goes.

4.8.2 Matrix-vector multiplication

The power of the matrix representation is that applying φ corresponds to matrix multiplication.

Represent a vector $x \in V$ by its coordinate column vector: if $x = \sum_{j=1}^n x_j v_j$, we write

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in k^n.$$

Similarly, $y = \varphi(x) \in W$ has coordinates $Y = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$ where $y = \sum_{i=1}^m y_i w_i$.

Proposition 4.8.1 Linear maps as matrix multiplication

With the notation above, $Y = AX$, i.e.,

$$y_i = \sum_{j=1}^n a_{ij} x_j.$$

Proof: Compute:

$$\begin{aligned} \varphi(x) &= \varphi\left(\sum_{j=1}^n x_j v_j\right) = \sum_{j=1}^n x_j \varphi(v_j) = \sum_{j=1}^n x_j \left(\sum_{i=1}^m a_{ij} w_i\right) \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} x_j\right) w_i. \end{aligned}$$

Reading off the coefficient of w_i , we get $y_i = \sum_{j=1}^n a_{ij} x_j$, which is exactly the i -th component of AX . $\square$

Note:-

As a memory aid: the isomorphism $k^n \xrightarrow{\sim} V$ given by a basis can be written symbolically as

$$(v_1 \ v_2 \ \cdots \ v_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \sum_{j=1}^n x_j v_j.$$

Similarly, φ acts as

$$\varphi((v_1 \cdots v_n)X) = (w_1 \cdots w_m)AX.$$

This notation makes the matrix formula $Y = AX$ visually clear.

4.8.3 The isomorphism $\text{Hom}(V, W) \cong \mathcal{M}_{m \times n}(k)$

Theorem 4.8.1 Linear maps correspond to matrices

Fix bases $\mathcal{B}$ for V and $\mathcal{C}$ for W . The map

$$\Phi : \text{Hom}(V, W) \rightarrow \mathcal{M}_{m \times n}(k), \quad \varphi \mapsto [\varphi]_{\mathcal{B}}^{\mathcal{C}}$$

sending a linear map to its matrix is an **isomorphism of vector spaces**.

Proof: We verify Φ is linear, injective, and surjective.

Linearity: Let $\varphi, \psi \in \text{Hom}(V, W)$ and $\lambda \in k$. The j -th column of $[\varphi + \psi]$ is the coordinate vector of $(\varphi + \psi)(v_j) = \varphi(v_j) + \psi(v_j)$, which is the sum of the j -th columns of $[\varphi]$ and $[\psi]$. Hence $[\varphi + \psi] = [\varphi] + [\psi]$. Similarly $[\lambda\varphi] = \lambda[\varphi]$.

Injectivity: If $[\varphi] = 0$ (the zero matrix), then every column is zero, meaning $\varphi(v_j) = 0$ for all basis vectors v_j . Since φ is linear and determined by its values on a basis, $\varphi = 0$.

Surjectivity: Given any matrix $A \in \mathcal{M}_{m \times n}(k)$, define $\varphi : V \rightarrow W$ by specifying $\varphi(v_j) = \sum_i a_{ij}w_i$ (the j -th column of A gives the coordinates of $\varphi(v_j)$). Extend linearly. Then $[\varphi] = A$. $\square$

Corollary 4.8.1 Dimension formula

$$\dim(\text{Hom}(V, W)) = \dim(\mathcal{M}_{m \times n}(k)) = mn.$$

Proof: The space $\mathcal{M}_{m \times n}(k)$ has the standard basis $\{E_{ij}\}$, where E_{ij} is the matrix with 1 in position (i, j) and 0 elsewhere. There are mn such matrices. $\square$

Note:-

This result says: **linear maps are matrices, once we choose coordinates**. The abstract theory of linear maps reduces to the concrete theory of matrix computations. However, the matrix depends on the choice of bases—changing bases changes the matrix (we will study this “change of basis” in the near future).

The interplay between the abstract (linear maps) and the concrete (matrices) is what makes linear algebra, well linear algebra (duh). Our goal is to understand how much of a linear map’s structure is intrinsic (independent of basis choice) versus coordinate-dependent.

4.9 Change of basis

What if we want to choose a different basis for V and/or W ? That is to say, what if we want to take vectors expressible in one basis to the same vectors expressed in another basis? Well, say that we are given two bases: (v_i) and (v'_i) in V . If given an expression of (v_i) we want to plug in and get an expression in terms of v'_i , we write this as

$$v'_j = \sum_i p_{ij}v_i.$$

So we get an $n \times m$ matrix P with entries (p_{ij}) , whose j^{th} column is the expression of v'_j in the basis (v_i) as a column vector. So symbolically, we can think of this as $(v'_i) = (v_i)P$. So $(v'_i)X' = (v_i)PX'$, i.e. the

element of V represented by a column vector X' in new basis (v'_i) is represented by $X = PX'$ in the old basis (v_i) . We can think of this more conceptually as P being a linear map—i.e. a matrix transformation—by $P = \mathcal{M}(\text{id}_V, (v'_i), (v_i))$. Breaking this down, we have the identity mapping taking the vector to itself (from one basis to another, but the same vector(s)); thus we are taking things from (v'_i) , here, to the targets of (v_i) . We can additionally do the same for W in reverse! Let $Q = \mathcal{M}(\text{id}_W, (w_i), (w'_i))$, i.e. $(w_i) = (w'_i)Q$. Hence:

$$\varphi((v'_i))X' = \varphi((v_i)PX') = (w_i)APX' = (w'_i)QAPX'.$$

That is to say, $\mathcal{M}(\varphi, (v'), (w')) = QAP$.

Note:-

Matrix of identity from (v_i) to (v'_i) would be P^{-1} which is the matrix such that pp^{-1} is the identity matrix.

If $V = W$ and we change basis using a single new basis (v'_i) for both domain and codomain, then a remarkable simplification occurs. Let $\varphi \in \text{End}(V) = \text{Hom}(V, V)$ be an **endomorphism** (a linear map from a space to itself). With $P = \mathcal{M}(\text{id}_V, (v'_i), (v_i))$ as above, and using the same basis change for both domain and codomain, we have $Q = P$ (since we're using (v_i) and (v'_i) for W as well). Thus the formula $\mathcal{M}(\varphi, (v'), (w')) = QAP$ becomes:

Theorem 4.9.1 Change of basis for endomorphisms

Let $\varphi \in \text{End}(V)$, let $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{B}' = (v'_1, \dots, v'_n)$ be two bases of V , and let $P = \mathcal{M}(\text{id}_V, \mathcal{B}', \mathcal{B})$ be the change-of-basis matrix. If $A = \mathcal{M}(\varphi, \mathcal{B})$ and $A' = \mathcal{M}(\varphi, \mathcal{B}')$, then

$$A' = P^{-1}AP.$$

Proof: We have established that $\mathcal{M}(\varphi, \mathcal{B}', \mathcal{B}') = QAP$ where $Q = \mathcal{M}(\text{id}_V, (v_i), (v'_i))$. But $Q = P^{-1}$ since applying P then Q (or vice versa) gives the identity transformation on coordinate vectors. Explicitly: if $X' = P^{-1}X$ converts old coordinates to new, and $X = PX'$ converts new to old, then composing identity maps gives $\mathcal{M}(\text{id}_V, (v_i), (v'_i)) = P^{-1}$.

Thus $A' = P^{-1}AP$. □

This formula tells us that the matrices representing the same linear operator in different bases are related by **conjugation** by an invertible matrix.

4.9.1 Similar matrices

Definition 4.9.1: Similar matrices

Two $n \times n$ matrices A and B over a field k are **similar** if there exists an invertible matrix $P \in \text{GL}_n(k)$ such that

$$B = P^{-1}AP.$$

We write $A \sim B$.

Proposition 4.9.1 Similarity is an equivalence relation

The similarity relation on $\mathcal{M}_{n \times n}(k)$ is an equivalence relation:

1. **Reflexive:** $A \sim A$ (take $P = I$).
2. **Symmetric:** If $A \sim B$, then $B \sim A$ (if $B = P^{-1}AP$, then $A = PBP^{-1} = (P^{-1})^{-1}B(P^{-1})$).
3. **Transitive:** If $A \sim B$ and $B \sim C$, then $A \sim C$ (if $B = P^{-1}AP$ and $C = Q^{-1}BQ$, then $C = Q^{-1}P^{-1}APQ = (PQ)^{-1}A(PQ)$).

As we can see, similarity classes of matrices correspond exactly to linear operators on k^n : two matrices are similar if and only if they represent the same linear operator in different bases.

Note:-

From the categorical perspective, similar matrices represent the same morphism in $\mathbf{Vect}_k$ —they are simply different coordinate descriptions of one intrinsic object. This is why similarity-invariant properties (trace, determinant, eigenvalues, rank) are so important: they are properties of the operator itself, not of any particular matrix representation.

4.9.2 Invariants under similarity

What properties of a matrix are preserved under similarity? These are called similarity invariants—they depend only on the underlying linear operator, not on the choice of basis.

Definition 4.9.2: Trace of a matrix

The trace of an $n \times n$ matrix $A = (a_{ij})$ is the sum of its diagonal entries:

$$\mathrm{Tr}(A) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \cdots + a_{nn}.$$

Lemma 4.9.1 Cyclic property of trace

For any matrices A and B such that both AB and BA are defined and square, we have $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$.

Proof: Let A be $m \times n$ and B be $n \times m$. Then

$$\mathrm{Tr}(AB) = \sum_{i=1}^m (AB)_{ii} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ji}.$$

Similarly,

$$\mathrm{Tr}(BA) = \sum_{j=1}^n (BA)_{jj} = \sum_{j=1}^n \sum_{i=1}^m b_{ji} a_{ij}.$$

These double sums are identical (just reorder the summation), so $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$. $\square$

Proposition 4.9.2 Trace is a similarity invariant

If $A \sim B$, then $\mathrm{Tr}(A) = \mathrm{Tr}(B)$. In particular, the trace of a linear operator $\varphi \in \mathrm{End}(V)$ is well-defined: $\mathrm{Tr}(\varphi) := \mathrm{Tr}(\mathcal{M}(\varphi, \mathcal{B}))$ for any basis $\mathcal{B}$.

Proof: If $B = P^{-1}AP$, then using the cyclic property:

$$\mathrm{Tr}(B) = \mathrm{Tr}(P^{-1}AP) = \mathrm{Tr}((P^{-1}A)P) = \mathrm{Tr}(P(P^{-1}A)) = \mathrm{Tr}((PP^{-1})A) = \mathrm{Tr}(A).$$

$\square$

Proposition 4.9.3 Determinant is a similarity invariant

If $A \sim B$, then $\det(A) = \det(B)$. Hence the determinant of a linear operator is well-defined.

Proof: If $B = P^{-1}AP$, then by the multiplicativity of determinants:

$$\det(B) = \det(P^{-1}AP) = \det(P^{-1}) \det(A) \det(P) = \frac{1}{\det(P)} \cdot \det(A) \cdot \det(P) = \det(A).$$

$\square$

Proposition 4.9.4 Eigenvalues are similarity invariants

If $A \sim B$, then A and B have the same eigenvalues (with the same algebraic multiplicities). In particular, the eigenvalues of a linear operator are intrinsic to the operator, not dependent on the choice of basis.

Proof: A scalar λ is an eigenvalue of A if and only if $\det(A - \lambda I) = 0$. If $B = P^{-1}AP$, then

$$B - \lambda I = P^{-1}AP - \lambda P^{-1}IP = P^{-1}(A - \lambda I)P.$$

Thus $B - \lambda I \sim A - \lambda I$, so $\det(B - \lambda I) = \det(A - \lambda I)$ by the previous result. Hence A and B have identical characteristic polynomials, and therefore the same eigenvalues with the same multiplicities. $\square$

Note:-

Rank is also a similarity invariant: $\text{rank}(P^{-1}AP) = \text{rank}(A)$ since multiplying by invertible matrices doesn't change rank. More sophisticated invariants include the **minimal polynomial**, the **Jordan normal form**, and the **rational canonical form**—these completely classify operators up to similarity over algebraically closed fields. We will encounter these when we study eigenvalues in depth.

Example 4.9.1

Let $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ over $\mathbb{R}$. Both have $\text{Tr} = 4$, $\det = 4$, and eigenvalue $\lambda = 2$ with multiplicity 2. Yet $A \not\sim B$! The reason: $B = 2I$, so $P^{-1}BP = P^{-1}(2I)P = 2I = B$ for any P . Since $A \neq B$, they cannot be similar.

This shows that trace, determinant, and eigenvalues are necessary but not sufficient conditions for similarity. The difference lies in the **geometric multiplicity** of the eigenvalue 2: for B , the eigenspace $E_2 = \ker(B - 2I) = \mathbb{R}^2$ has dimension 2; for A , $E_2 = \ker(A - 2I) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \text{Span}\{(1, 0)\}$ has dimension 1. Geometric multiplicity is also a similarity invariant!

Question 2

1. Let $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear map with matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ with respect to the standard basis. Find the matrix of φ with respect to the basis $\mathcal{B}' = \{(1, 1), (1, -1)\}$.
2. Prove that an $n \times n$ matrix A is similar to a diagonal matrix if and only if there exists a basis of k^n consisting of eigenvectors of A .
3. Show that $\text{Tr}(AB) = \text{Tr}(BA)$ but in general $\det(A + B) \neq \det(A) + \det(B)$. Give a counterexample.

Solution

(1) First, find the change-of-basis matrix P from $\mathcal{B}'$ to the standard basis $\mathcal{E}$. The columns of P are the vectors of $\mathcal{B}'$ expressed in $\mathcal{E}$:

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

$$\text{Then } P^{-1} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}.$$

The matrix of φ with respect to $\mathcal{B}'$ is:

$$A' = P^{-1}AP = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

$$\text{Computing } AP \text{ first: } AP = \begin{pmatrix} 3 & -1 \\ 7 & -1 \end{pmatrix}.$$

Then $P^{-1}(AP) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 7 & -1 \end{pmatrix} = \begin{pmatrix} 5 & -1 \\ -2 & 0 \end{pmatrix}$.

So $A' = \begin{pmatrix} 5 & -1 \\ -2 & 0 \end{pmatrix}$. We can verify: $\text{Tr}(A') = 5 = \text{Tr}(A)$ and $\det(A') = 0 - 2 = -2 = \det(A)$. ✓

(2) ($\Rightarrow$) Suppose A is similar to a diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, so $D = P^{-1}AP$ for some invertible P . Let v_i be the i -th column of P . Then $AP = PD$, which gives $Av_i = \lambda_i v_i$. So each column of P is an eigenvector of A . Since P is invertible, its columns form a basis of k^n .

($\Leftarrow$) Suppose $\{v_1, \dots, v_n\}$ is a basis of eigenvectors with $Av_i = \lambda_i v_i$. Let P be the matrix with columns v_i . Then $AP = PD$ where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, so $D = P^{-1}AP$.

(3) We proved $\text{Tr}(AB) = \text{Tr}(BA)$ above. For the determinant counterexample:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then $\det(A) = 0$, $\det(B) = 0$, but $A + B = I$, so $\det(A + B) = 1 \neq 0 = \det(A) + \det(B)$.

4.10 Kernel and Image

Every linear map carries two intrinsic subspaces that reveal its fundamental structure: the kernel (what gets annihilated) and the image (what gets hit). These are the linear-algebraic analogues of concepts we've seen before: the kernel of a group homomorphism and the image of any function. The intermingling betwixt (what a fun word) kernel and image is governed by the Rank-Nullity Theorem.

Note:-

From the categorical viewpoint, kernel and image are two ways to “measure” a morphism. The kernel measures the failure of injectivity—how much information the map destroys. The image measures the failure of surjectivity—how much of the codomain the map misses. The Rank-Nullity Theorem says these two failures are complementary: the more a linear map kills, the less it can produce, and vice versa. A linear map $\varphi : V \rightarrow W$ is completely determined by: (1) which subspace it kills (the kernel), and (2) which subspace it hits (the image). Once you know these, there's essentially only one way to fill in the rest—the map restricted to any complement of the kernel is an isomorphism onto the image. This is different from nonlinear maps, where knowing the zero set tells you almost nothing about the map's behavior elsewhere.

4.10.1 The Kernel (Null Space)

Definition 4.10.1: Kernel of a linear map

Let $\varphi : V \rightarrow W$ be a linear map. The kernel (or null space) of φ is

$$\ker(\varphi) = \{v \in V : \varphi(v) = 0_W\}.$$

This is the set of all vectors that φ maps to zero.

Note:-

The terminology “kernel” comes from algebra (group homomorphisms), while “null space” is more common in applied linear algebra and matrix theory. We use both interchangeably. In categorical language, the kernel is the equalizer of φ and the zero map.

Example 4.10.1 (Examples of kernels)

1. The zero map $0 : V \rightarrow W$ has $\ker(0) = V$.
2. The identity map $\text{id}_V : V \rightarrow V$ has $\ker(\text{id}_V) = \{0\}$.
3. Let $\varphi : \mathbb{R}^3 \rightarrow \mathbb{R}$ be defined by $\varphi(x, y, z) = x + 2y + 3z$. Then $\ker(\varphi) = \{(x, y, z) \in \mathbb{R}^3 : x + 2y + 3z = 0\}$, a plane through the origin.
4. Let $D : \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$ be differentiation, $Dp = p'$. Then $\ker(D)$ is the set of constant polynomials.
5. Let $T : \mathcal{M}_{n \times n}(k) \rightarrow k$ be the trace map, $T(A) = \text{Tr}(A)$. Then $\ker(T)$ is the space of traceless matrices.

Proposition 4.10.1 The kernel is a subspace

Let $\varphi : V \rightarrow W$ be a linear map. Then $\ker(\varphi)$ is a subspace of V .

Proof: We verify the subspace criteria:

1. **Contains zero:** Since $\varphi(0_V) = 0_W$ (every linear map preserves zero), we have $0_V \in \ker(\varphi)$.
2. **Closed under addition:** If $u, v \in \ker(\varphi)$, then $\varphi(u+v) = \varphi(u) + \varphi(v) = 0 + 0 = 0$, so $u+v \in \ker(\varphi)$.
3. **Closed under scalar multiplication:** If $v \in \ker(\varphi)$ and $\lambda \in k$, then $\varphi(\lambda v) = \lambda \varphi(v) = \lambda \cdot 0 = 0$, so $\lambda v \in \ker(\varphi)$.

□

Note:-

Compare this with group theory: the kernel of a group homomorphism $f : G \rightarrow H$ is a normal subgroup of G . For linear maps, every subspace is “normal” in the appropriate sense—vector spaces are abelian, so there’s no distinction between left and right cosets.

The kernel provides a clean characterization of injectivity:

Theorem 4.10.1 Injectivity via the kernel

Let $\varphi : V \rightarrow W$ be a linear map. Then

$$\varphi \text{ is injective} \iff \ker(\varphi) = \{0\}.$$

Proof: ($\Rightarrow$) Suppose φ is injective. Let $v \in \ker(\varphi)$. Then $\varphi(v) = 0 = \varphi(0)$. Since φ is injective, $v = 0$. Thus $\ker(\varphi) \subseteq \{0\}$, and since $0 \in \ker(\varphi)$ always, we have $\ker(\varphi) = \{0\}$.

($\Leftarrow$) Suppose $\ker(\varphi) = \{0\}$. Let $u, v \in V$ with $\varphi(u) = \varphi(v)$. Then

$$\varphi(u - v) = \varphi(u) - \varphi(v) = 0,$$

so $u - v \in \ker(\varphi) = \{0\}$. Thus $u - v = 0$, i.e., $u = v$. Hence φ is injective.

□

Corollary 4.10.1

An injective linear map from V to W embeds V as a subspace of W . More precisely, if $\varphi : V \rightarrow W$ is injective, then $V \cong \text{Im}(\varphi) \subseteq W$.

Note:-

Here the kernel has a role more than merely verifying injectivity—it measures the failure of injectivity in a precise sense. Two vectors $u, v \in V$ have the same image under φ if and only if $u - v \in \ker(\varphi)$. Thus the fibers of φ (preimages of points) are exactly the cosets $v + \ker(\varphi)$. The map φ is injective precisely when these cosets are singletons, i.e., when $\ker(\varphi) = \{0\}$. This is why we can “mod out by the kernel” to obtain an injective map—the quotient $V/\ker(\varphi)$ collapses each fiber to a point.

4.10.2 The Image (Range)

Definition 4.10.2: Image of a linear map

Let $\varphi : V \rightarrow W$ be a linear map. The **image** (or **range**) of φ is

$$\text{Im}(\varphi) = \{\varphi(v) : v \in V\} = \{w \in W : w = \varphi(v) \text{ for some } v \in V\}.$$

This is the set of all vectors that φ “hits.”

Example 4.10.2 (Examples of images)

1. The zero map $0 : V \rightarrow W$ has $\text{Im}(0) = \{0_W\}$.
2. The identity map $\text{id}_V : V \rightarrow V$ has $\text{Im}(\text{id}_V) = V$.
3. Let $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be projection $\pi(x, y, z) = (x, y)$. Then $\text{Im}(\pi) = \mathbb{R}^2$.
4. Let $D : \mathcal{P}_n(\mathbb{R}) \rightarrow \mathcal{P}_{n-1}(\mathbb{R})$ be differentiation. Then $\text{Im}(D) = \mathcal{P}_{n-1}(\mathbb{R})$ (every polynomial of degree $\leq n-1$ is the derivative of some polynomial of degree $\leq n$).
5. Let $A \in \mathcal{M}_{m \times n}(k)$ define $L_A : k^n \rightarrow k^m$. Then $\text{Im}(L_A) = \text{ColSpace}(A)$, the column space of A .

Proposition 4.10.2 The image is a subspace

Let $\varphi : V \rightarrow W$ be a linear map. Then $\text{Im}(\varphi)$ is a subspace of W .

Proof: We verify the subspace criteria:

1. **Contains zero:** $\varphi(0_V) = 0_W$, so $0_W \in \text{Im}(\varphi)$.
2. **Closed under addition:** If $w_1, w_2 \in \text{Im}(\varphi)$, then $w_1 = \varphi(v_1)$ and $w_2 = \varphi(v_2)$ for some $v_1, v_2 \in V$. Thus $w_1 + w_2 = \varphi(v_1) + \varphi(v_2) = \varphi(v_1 + v_2) \in \text{Im}(\varphi)$.
3. **Closed under scalar multiplication:** If $w \in \text{Im}(\varphi)$ and $\lambda \in k$, then $w = \varphi(v)$ for some $v \in V$, so $\lambda w = \lambda \varphi(v) = \varphi(\lambda v) \in \text{Im}(\varphi)$.

□

Definition 4.10.3: Surjectivity

A linear map $\varphi : V \rightarrow W$ is **surjective** (or **onto**) if $\text{Im}(\varphi) = W$, i.e., every element of W is hit by some element of V .

Proposition 4.10.3 Characterization of surjectivity

Let $\varphi : V \rightarrow W$ be a linear map. The following are equivalent:

1. φ is surjective.
2. $\text{Im}(\varphi) = W$.

3. If $\mathcal{B}$ spans V , then $\varphi(\mathcal{B}) = \{\varphi(v) : v \in \mathcal{B}\}$ spans W .

4.10.3 The Rank-Nullity Theorem

We now come to the central result connecting kernel and image. This theorem quantifies the fundamental trade-off in linear algebra: the more a map “kills,” the less it can “produce.”

Definition 4.10.4: Rank and nullity

Let $\varphi : V \rightarrow W$ be a linear map between finite-dimensional vector spaces. The **rank** of φ is

$$\text{rank}(\varphi) = \dim(\text{Im}(\varphi)),$$

and the **nullity** of φ is

$$\text{null}(\varphi) = \dim(\ker(\varphi)).$$

For a matrix A , we define $\text{rank}(A)$ and $\text{null}(A)$ to be the rank and nullity of the corresponding linear map.

Theorem 4.10.2 The Rank-Nullity Theorem (Fundamental Theorem of Linear Maps)

Let $\varphi : V \rightarrow W$ be a linear map with V finite-dimensional. Then $\text{Im}(\varphi)$ is finite-dimensional and

$$\dim(V) = \dim(\ker(\varphi)) + \dim(\text{Im}(\varphi)).$$

Equivalently: $\dim(V) = \text{null}(\varphi) + \text{rank}(\varphi)$.

Proof: Let $\{u_1, \dots, u_m\}$ be a basis for $\ker(\varphi)$. By the basis extension theorem, we can extend this to a basis

$$\{u_1, \dots, u_m, v_1, \dots, v_r\}$$

of V , where $\dim(V) = m + r$.

Claim: $\{\varphi(v_1), \dots, \varphi(v_r)\}$ is a basis for $\text{Im}(\varphi)$.

Spanning: Let $w \in \text{Im}(\varphi)$, say $w = \varphi(x)$ for some $x \in V$. Write

$$x = a_1 u_1 + \dots + a_m u_m + b_1 v_1 + \dots + b_r v_r.$$

Applying φ :

$$w = \varphi(x) = a_1 \varphi(u_1) + \dots + a_m \varphi(u_m) + b_1 \varphi(v_1) + \dots + b_r \varphi(v_r).$$

Since each $u_i \in \ker(\varphi)$, we have $\varphi(u_i) = 0$, so

$$w = b_1 \varphi(v_1) + \dots + b_r \varphi(v_r).$$

Thus $\{\varphi(v_1), \dots, \varphi(v_r)\}$ spans $\text{Im}(\varphi)$.

Linear independence: Suppose

$$c_1 \varphi(v_1) + \dots + c_r \varphi(v_r) = 0.$$

Then $\varphi(c_1 v_1 + \dots + c_r v_r) = 0$, so $c_1 v_1 + \dots + c_r v_r \in \ker(\varphi)$. Since $\{u_1, \dots, u_m\}$ is a basis for $\ker(\varphi)$, we can write

$$c_1 v_1 + \dots + c_r v_r = d_1 u_1 + \dots + d_m u_m$$

for some scalars d_j . Rearranging:

$$d_1 u_1 + \dots + d_m u_m - c_1 v_1 - \dots - c_r v_r = 0.$$

Since $\{u_1, \dots, u_m, v_1, \dots, v_r\}$ is linearly independent (it's a basis of V), all coefficients are zero. In particular, $c_1 = \dots = c_r = 0$.

Thus $\{\varphi(v_1), \dots, \varphi(v_r)\}$ is a basis for $\text{Im}(\varphi)$, so $\dim(\text{Im}(\varphi)) = r$. We conclude:

$$\dim(V) = m + r = \dim(\ker(\varphi)) + \dim(\text{Im}(\varphi)).$$

□

Note:-

The Rank-Nullity Theorem is the linear algebra version of the **First Isomorphism Theorem** from group theory. Recall that for a group homomorphism $f : G \rightarrow H$, we have $G/\ker(f) \cong \text{Im}(f)$. The linear version states $V/\ker(\varphi) \cong \text{Im}(\varphi)$, and “counting dimensions” gives $\dim(V) - \dim(\ker(\varphi)) = \dim(\text{Im}(\varphi))$.

4.10.4 Consequences of the Rank-Nullity Theorem

The Rank-Nullity Theorem has powerful implications for understanding linear maps.

Corollary 4.10.2 Dimension constraints on injectivity

Let $\varphi : V \rightarrow W$ be a linear map with $\dim(V) = n$ and $\dim(W) = m$.

1. If $n > m$, then φ cannot be injective.
2. If $n < m$, then φ cannot be surjective.

Proof: (1) If φ were injective, then $\dim(\ker(\varphi)) = 0$, so by Rank-Nullity:

$$\dim(\text{Im}(\varphi)) = \dim(V) = n > m = \dim(W).$$

But $\text{Im}(\varphi) \subseteq W$, so $\dim(\text{Im}(\varphi)) \leq \dim(W) = m$. Contradiction.

(2) If φ were surjective, then $\dim(\text{Im}(\varphi)) = \dim(W) = m$, so by Rank-Nullity:

$$\dim(\ker(\varphi)) = \dim(V) - \dim(\text{Im}(\varphi)) = n - m < 0.$$

But dimensions are non-negative. Contradiction.

□

Corollary 4.10.3 Equal dimensions: bijectivity criteria

Let $\varphi : V \rightarrow W$ be a linear map with $\dim(V) = \dim(W) = n$. The following are equivalent:

1. φ is injective.
2. φ is surjective.
3. φ is bijective (an isomorphism).
4. $\ker(\varphi) = \{0\}$.
5. $\text{Im}(\varphi) = W$.
6. $\text{rank}(\varphi) = n$.

Proof: By Rank-Nullity, $\text{null}(\varphi) + \text{rank}(\varphi) = n$. Since $\text{rank}(\varphi) \leq \dim(W) = n$:

- φ injective $\iff \text{null}(\varphi) = 0 \iff \text{rank}(\varphi) = n$.
- φ surjective $\iff \text{rank}(\varphi) = n \iff \text{null}(\varphi) = 0$.

Thus injectivity, surjectivity, and bijectivity are equivalent when dimensions match.

□

Note:-

For linear maps between equal-dimensional spaces, “one-to-one” and “onto” are the same condition. This fails for general functions and even for linear maps between infinite-dimensional spaces. Consider the **right shift** $R : k^\infty \rightarrow k^\infty$ defined by $R(x_1, x_2, x_3, \dots) = (0, x_1, x_2, \dots)$. This is injective (kernel is trivial) but not surjective (the sequence $(1, 0, 0, \dots)$ is not in the image). Conversely, the **left shift** $L(x_1, x_2, x_3, \dots) = (x_2, x_3, \dots)$ is surjective but not injective. In infinite dimensions, there’s “room” for one without the other. The finite-dimensional equivalence is a manifestation of the pigeonhole principle applied to basis vectors.

Corollary 4.10.4 Existence of right/left inverses

Let $\varphi : V \rightarrow W$ be a linear map between finite-dimensional spaces.

1. φ has a left inverse if and only if φ is injective.
2. φ has a right inverse if and only if φ is surjective.

4.10.5 Matrix Interpretation

For a matrix $A \in \mathcal{M}_{m \times n}(k)$, the associated linear map is $L_A : k^n \rightarrow k^m$ given by $L_A(x) = Ax$. The Rank-Nullity Theorem becomes:

Proposition 4.10.4 Rank-Nullity for matrices

For $A \in \mathcal{M}_{m \times n}(k)$:

1. $\ker(L_A) = \{x \in k^n : Ax = 0\}$ is the null space of A .
2. $\text{Im}(L_A) = \{Ax : x \in k^n\}$ is the column space of A (the span of the columns).
3. $\text{null}(A) + \text{rank}(A) = n$ (the number of columns).

Note:-

The rank of a matrix equals the number of linearly independent columns, which equals the number of linearly independent rows. The column space has dimension equal to the rank, and the null space has dimension equal to $n - \text{rank}(A)$.

Looking ahead: when we study eigenvalues, the kernel will reappear in a crucial role. For a linear operator $T : V \rightarrow V$ and scalar λ , the **eigenspace** $E_\lambda = \ker(T - \lambda I)$ is precisely the kernel of the shifted operator. The dimension of this eigenspace (the geometric multiplicity of λ) measures “how degenerate” the eigenvalue is. The interplay between algebraic and geometric multiplicities—and the failure of these to match—is what makes Jordan normal form necessary.

Example 4.10.3 (Computing kernel and image)

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 1 & 3 & 5 \end{pmatrix}$ over $\mathbb{R}$.

Finding $\ker(A)$: Solve $Ax = 0$. Row reduce:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 1 & 3 & 5 \end{pmatrix} \xrightarrow{R_3 - R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_3 - R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Back-substituting: $x_2 = -2x_3$, $x_1 = -2x_2 - 3x_3 = 4x_3 - 3x_3 = x_3$. So

$$\ker(A) = \text{Span}\{(1, -2, 1)^T\}, \quad \text{null}(A) = 1.$$

Finding $\text{Im}(A)$: The non-zero rows in RREF correspond to pivot columns. Columns 1 and 2 are pivot columns, so

$$\text{Im}(A) = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \right\}, \quad \text{rank}(A) = 2.$$

Verification: $\text{null}(A) + \text{rank}(A) = 1 + 2 = 3 = n$. ✓

Question 3

1. Let $D : \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_3(\mathbb{R})$ be the differentiation operator. Find $\ker(D)$, $\text{Im}(D)$, $\text{null}(D)$, and $\text{rank}(D)$. Verify the Rank-Nullity Theorem.
2. Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be defined by $T(x_1, x_2, x_3, x_4) = (x_1 - x_2, x_2 - x_3, x_3 - x_4)$. Determine whether T is injective, surjective, or neither.
3. Prove that if $\varphi : V \rightarrow V$ is a linear operator on a finite-dimensional space with $\varphi^2 = \varphi$, then $V = \ker(\varphi) \oplus \text{Im}(\varphi)$.
4. Let V and W be finite-dimensional vector spaces with $\dim(V) = \dim(W)$. Prove that a linear map $\varphi : V \rightarrow W$ is injective if and only if for every linearly independent list in V , its image under φ is linearly independent in W .

Solution

(1) The space $\mathcal{P}_3(\mathbb{R})$ has basis $\{1, x, x^2, x^3\}$, so $\dim(\mathcal{P}_3(\mathbb{R})) = 4$.

Kernel: $Dp = 0$ if and only if p is a constant polynomial. So $\ker(D) = \text{Span}\{1\}$, which are the constant functions. Thus $\text{null}(D) = 1$.

Image: Differentiating: $D(1) = 0$, $D(x) = 1$, $D(x^2) = 2x$, $D(x^3) = 3x^2$. The non-zero outputs span $\text{Span}\{1, x, x^2\} = \mathcal{P}_2(\mathbb{R})$. Thus $\text{Im}(D) = \mathcal{P}_2(\mathbb{R})$ and $\text{rank}(D) = 3$.

Verification: $\text{null}(D) + \text{rank}(D) = 1 + 3 = 4 = \dim(\mathcal{P}_3(\mathbb{R}))$. ✓

(2) **Injectivity:** Find $\ker(T)$. Solve $T(x_1, x_2, x_3, x_4) = (0, 0, 0)$:

$$x_1 - x_2 = 0, \quad x_2 - x_3 = 0, \quad x_3 - x_4 = 0.$$

This gives $x_1 = x_2 = x_3 = x_4$. So $\ker(T) = \{(t, t, t, t) : t \in \mathbb{R}\} = \text{Span}\{(1, 1, 1, 1)\}$.

Since $\ker(T) \neq \{0\}$, T is **not injective**.

Surjectivity: We have $\text{null}(T) = 1$, so $\text{rank}(T) = 4 - 1 = 3 = \dim(\mathbb{R}^3)$. Thus $\text{Im}(T) = \mathbb{R}^3$, and T is **surjective**.

(3) We show $V = \ker(\varphi) + \text{Im}(\varphi)$ and $\ker(\varphi) \cap \text{Im}(\varphi) = \{0\}$.

Sum equals V: Let $v \in V$. Write $v = (v - \varphi(v)) + \varphi(v)$. Note that $\varphi(v) \in \text{Im}(\varphi)$. Also:

$$\varphi(v - \varphi(v)) = \varphi(v) - \varphi^2(v) = \varphi(v) - \varphi(v) = 0,$$

so $v - \varphi(v) \in \ker(\varphi)$. Thus $v \in \ker(\varphi) + \text{Im}(\varphi)$.

Intersection is trivial: Let $w \in \ker(\varphi) \cap \text{Im}(\varphi)$. Then $\varphi(w) = 0$ and $w = \varphi(u)$ for some $u \in V$. So:

$$w = \varphi(u) = \varphi^2(u) = \varphi(\varphi(u)) = \varphi(w) = 0.$$

Thus $V = \ker(\varphi) \oplus \text{Im}(\varphi)$.

(4) ($\Rightarrow$) Suppose φ is injective and $\{v_1, \dots, v_m\}$ is linearly independent. Suppose $\sum c_i \varphi(v_i) = 0$. Then $\varphi(\sum c_i v_i) = 0$, so $\sum c_i v_i \in \ker(\varphi) = \{0\}$. Thus $\sum c_i v_i = 0$, and by linear independence of the v_i , all $c_i = 0$. Hence $\{\varphi(v_1), \dots, \varphi(v_m)\}$ is linearly independent.

($\Leftarrow$) Suppose φ maps every linearly independent set to a linearly independent set. If $v \neq 0$, then $\{v\}$ is linearly independent, so $\{\varphi(v)\}$ is linearly independent, meaning $\varphi(v) \neq 0$. Thus $\ker(\varphi) = \{0\}$, so φ is injective.

4.11 Duality

The dual space of a vector space V consists of all linear maps from V to the base field k —these are called linear functionals. While this might seem like a minor variation on Hom -spaces, duality reveals useful structure: every basis has a “dual basis,” every linear map has a “dual map,” and the matrix of the dual map is the transpose of the original. In more advanced contexts (differential geometry, functional analysis, representation theory), duality becomes indispensable.

Note:-

The dual space construction is the first example of a **contravariant functor** you’ll encounter. When we pass from V to V' , linear maps $T : V \rightarrow W$ don’t induce maps $V' \rightarrow W'$ —instead, they induce maps $W' \rightarrow V'$, going the “wrong way.” This reversal of arrows is the hallmark of contravariance. In category theory, the dual space functor $(-)' : \mathbf{Vect}_k^{\text{op}} \rightarrow \mathbf{Vect}_k$ is representable: $V' = \text{Hom}(V, k)$ is “the functor represented by k .”

4.11.1 Linear Functionals and the Dual Space

Definition 4.11.1: Linear functional

A linear functional on a vector space V over a field k is a linear map $\ell : V \rightarrow k$. In other words, a linear functional is an element of $\text{Hom}(V, k)$.

Example 4.11.1 (Examples of linear functionals)

1. **Coordinate projections:** On k^n , define $\pi_j(x_1, \dots, x_n) = x_j$. Each π_j is a linear functional.
2. **Weighted sums:** Fix $(c_1, \dots, c_n) \in k^n$. Define $\ell(x_1, \dots, x_n) = c_1x_1 + \dots + c_nx_n$. This is a linear functional on k^n , and every linear functional on k^n has this form.
3. **Evaluation:** On $\mathcal{P}(\mathbb{R})$, define $\text{ev}_a(p) = p(a)$ for fixed $a \in \mathbb{R}$. This is a linear functional.
4. **Integration:** On $\mathcal{P}(\mathbb{R})$, define $\ell(p) = \int_0^1 p(x) dx$. Linearity of the integral makes this a linear functional.
5. **Trace:** On $\mathcal{M}_{n \times n}(k)$, the trace $\text{Tr}(A) = \sum_{i=1}^n a_{ii}$ is a linear functional.

Definition 4.11.2: Dual space

The dual space of V , denoted V^* or V' , is the vector space of all linear functionals on V :

$$V^* = \text{Hom}(V, k).$$

This is a vector space under pointwise addition and scalar multiplication: $(\ell_1 + \ell_2)(v) = \ell_1(v) + \ell_2(v)$ and $(\lambda\ell)(v) = \lambda \cdot \ell(v)$.

Note:-

We use V^* notation (common in differential geometry and physics) interchangeably with V' (common in algebra). Both mean the same thing. Some texts reserve V^* for the continuous dual in functional analysis, but for finite-dimensional spaces this distinction is irrelevant.

Proposition 4.11.1 Dimension of the dual space

If V is finite-dimensional with $\dim(V) = n$, then $\dim(V^*) = n$. Thus $V^* \cong V$ as vector spaces (non-canonically).

Proof: By our earlier result on Hom-spaces:

$$\dim(V^*) = \dim(\text{Hom}(V, k)) = \dim(V) \cdot \dim(k) = n \cdot 1 = n.$$

□

Note:-

The isomorphism $V \cong V^*$ is non-canonical—it depends on a choice of basis. There’s no “natural” way to identify vectors with linear functionals without extra structure. However, the double dual $V^{**} = (V^*)^*$ does have a canonical isomorphism with V , given by evaluation: $v \mapsto (\ell \mapsto \ell(v))$.

4.11.2 The Dual Basis

Given a basis of V , there’s a natural associated basis of V^* .

Definition 4.11.3: Dual basis

Let $\mathcal{B} = \{v_1, \dots, v_n\}$ be a basis of V . The dual basis is the list $\mathcal{B}^* = \{\varphi_1, \dots, \varphi_n\}$ of linear functionals in V^* defined by

$$\varphi_j(v_i) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

In other words, φ_j “picks out” the j -th coordinate: if $v = c_1v_1 + \dots + c_nv_n$, then $\varphi_j(v) = c_j$.

Proposition 4.11.2 Dual basis is a basis

The dual basis $\{\varphi_1, \dots, \varphi_n\}$ is indeed a basis of V^* .

Proof: **Linear independence:** Suppose $a_1\varphi_1 + \dots + a_n\varphi_n = 0$ (the zero functional). Evaluating at v_k :

$$0 = (a_1\varphi_1 + \dots + a_n\varphi_n)(v_k) = a_1\varphi_1(v_k) + \dots + a_n\varphi_n(v_k) = a_k.$$

So all $a_k = 0$.

Spanning: Since we have n linearly independent vectors in V^* and $\dim(V^*) = n$, they form a basis. □

Proposition 4.11.3 Dual basis extracts coordinates

Let $\{v_1, \dots, v_n\}$ be a basis of V with dual basis $\{\varphi_1, \dots, \varphi_n\}$. For any $v \in V$:

$$v = \varphi_1(v)v_1 + \varphi_2(v)v_2 + \dots + \varphi_n(v)v_n.$$

That is, $\varphi_j(v)$ is the j -th coordinate of v with respect to the basis $\{v_i\}$.

Proof: Write $v = c_1v_1 + \dots + c_nv_n$. Applying φ_j :

$$\varphi_j(v) = c_1\varphi_j(v_1) + \dots + c_n\varphi_j(v_n) = c_j.$$

Substituting back: $v = \varphi_1(v)v_1 + \dots + \varphi_n(v)v_n$. □

Example 4.11.2 (Dual basis of the standard basis)

Let $\{e_1, \dots, e_n\}$ be the standard basis of k^n . The dual basis consists of the coordinate projections:

$$\varphi_j(x_1, \dots, x_n) = x_j.$$

These are the “row vectors” $\varphi_j = (0, \dots, 0, 1, 0, \dots, 0)$ (with 1 in position j).

4.11.3 The Dual Map

Just as a linear map $T : V \rightarrow W$ induces a map on Hom-spaces, it induces a map on dual spaces—but in the opposite direction.

Definition 4.11.4: Dual map (transpose)

Let $T : V \rightarrow W$ be a linear map. The **dual map** (or **transpose**) is the linear map $T^* : W^* \rightarrow V^*$ defined by

$$T^*(\psi) = \psi \circ T$$

for each $\psi \in W^*$. In other words, $T^*(\psi)$ is the linear functional on V given by $v \mapsto \psi(T(v))$.

Note:-

The dual map goes “backwards”: $T : V \rightarrow W$ yields $T^* : W^* \rightarrow V^*$. This is because we’re precomposing with T . If you think of linear functionals as “measurements,” then $T^*(\psi)$ is the measurement you get by first applying T , then measuring with ψ . The reversal of direction is the essence of contravariance.

Proposition 4.11.4 The dual map is linear

For $T : V \rightarrow W$, the dual map $T^* : W^* \rightarrow V^*$ is indeed a linear map.

Proof: For $\psi_1, \psi_2 \in W^*$ and $\lambda \in k$:

$$\begin{aligned} T^*(\psi_1 + \psi_2) &= (\psi_1 + \psi_2) \circ T = \psi_1 \circ T + \psi_2 \circ T = T^*(\psi_1) + T^*(\psi_2), \\ T^*(\lambda\psi) &= (\lambda\psi) \circ T = \lambda(\psi \circ T) = \lambda T^*(\psi). \end{aligned}$$

□

Proposition 4.11.5 Algebraic properties of the dual map

Let $S : V \rightarrow W$ and $T : W \rightarrow U$ be linear maps.

1. $(S + T)^* = S^* + T^*$ (when both are defined).
2. $(\lambda T)^* = \lambda T^*$ for $\lambda \in k$.
3. $(T \circ S)^* = S^* \circ T^*$ (order reverses!).
4. $(\text{id}_V)^* = \text{id}_{V^*}$.

Proof: (3) is the key property. For $\psi \in U^*$:

$$(T \circ S)^*(\psi) = \psi \circ (T \circ S) = (\psi \circ T) \circ S = S^*(\psi \circ T) = S^*(T^*(\psi)) = (S^* \circ T^*)(\psi).$$

The other properties are similar.

□

Note:-

Property (3) says the dual is a contravariant functor: it reverses the order of composition. Compare with the transpose of matrices: $(AB)^T = B^T A^T$. This is not a coincidence—the matrix of the dual map is the transpose!

4.11.4 Matrix of the Dual Map

The connection between dual maps and matrix transposes clarifies why the transpose operation is natural.

Theorem 4.11.1 Matrix of the dual map is the transpose

Let $T : V \rightarrow W$ be a linear map between finite-dimensional spaces. Fix bases $\mathcal{B} = \{v_1, \dots, v_n\}$ of V and $\mathcal{C} = \{w_1, \dots, w_m\}$ of W , with dual bases $\mathcal{B}^* = \{\varphi_1, \dots, \varphi_n\}$ and $\mathcal{C}^* = \{\psi_1, \dots, \psi_m\}$. If

$$A = \mathcal{M}(T; \mathcal{B}, \mathcal{C}) \in \mathcal{M}_{m \times n}(k)$$

is the matrix of T , then the matrix of T^* with respect to the dual bases is the transpose:

$$\mathcal{M}(T^*; \mathcal{C}^*, \mathcal{B}^*) = A^T \in \mathcal{M}_{n \times m}(k).$$

Proof: We need to show that $T^*(\psi_j) = \sum_{i=1}^n A_{ji} \varphi_i$ for each j .

By definition of the matrix A , we have $T(v_k) = \sum_{i=1}^m A_{ik} w_i$. Now compute:

$$T^*(\psi_j)(v_k) = \psi_j(T(v_k)) = \psi_j\left(\sum_{i=1}^m A_{ik} w_i\right) = \sum_{i=1}^m A_{ik} \psi_j(w_i) = A_{jk}.$$

On the other hand, if $T^*(\psi_j) = \sum_{i=1}^n c_i \varphi_i$, then

$$T^*(\psi_j)(v_k) = \sum_{i=1}^n c_i \varphi_i(v_k) = c_k.$$

Comparing: $c_k = A_{jk}$. Thus $T^*(\psi_j) = \sum_{k=1}^n A_{jk} \varphi_k$, which means the j -th column of $\mathcal{M}(T^*)$ is the j -th row of A . Hence $\mathcal{M}(T^*) = A^T$. $\square$

Note:-

This theorem explains why the transpose is natural: it's not just a matrix operation, it's the coordinate representation of the dual map. The reversal of dimensions ($m \times n$ becomes $n \times m$) reflects the contravariance of the dual construction.

Corollary 4.11.1 Rank of transpose equals rank

For any matrix A , $\text{rank}(A^T) = \text{rank}(A)$.

Proof: The rank of A is $\dim(\text{Im}(T))$ where T is the associated linear map. By a result we'll see shortly, $\dim(\text{Im}(T^*)) = \dim(\text{Im}(T))$. Since the matrix of T^* is A^T , we get $\text{rank}(A^T) = \text{rank}(A)$. $\square$

4.11.5 Kernel and Image of the Dual Map

The dual map interchanges the roles of kernel and image in a precise way.

Proposition 4.11.6 Relationships between T and T^*

Let $T : V \rightarrow W$ be a linear map between finite-dimensional spaces.

1. $\ker(T^*) = (\text{Im}(T))^\circ$, the annihilator of the image.
2. $\text{Im}(T^*) = (\ker(T))^\circ$, the annihilator of the kernel.
3. $\dim(\ker(T^*)) = \dim(W) - \text{rank}(T)$.
4. $\dim(\text{Im}(T^*)) = \text{rank}(T)$.

Here, for a subspace $U \subseteq V$, the annihilator is $U^\circ = \{\ell \in V^* : \ell(u) = 0 \text{ for all } u \in U\}$.

Note:-

The annihilator U° consists of all linear functionals that vanish on U . If $\dim(V) = n$ and $\dim(U) = k$, then $\dim(U^\circ) = n - k$. The annihilator provides a duality between subspaces of V and subspaces of V^* : bigger subspaces have smaller annihilators.

Question 4

1. Let $V = \mathbb{R}^3$ with standard basis $\{e_1, e_2, e_3\}$. Write down the dual basis explicitly as functions $\mathbb{R}^3 \rightarrow \mathbb{R}$.
2. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, y + z)$. Find the matrix of T with respect to the standard bases, then find the matrix of T^* with respect to the dual bases.
3. Prove that if $T : V \rightarrow W$ is surjective, then $T^* : W^* \rightarrow V^*$ is injective.
4. Prove that if V is finite-dimensional, the map $\Phi : V \rightarrow V^{**}$ defined by $\Phi(v)(\ell) = \ell(v)$ is an isomorphism.

Solution

(1) The dual basis consists of the coordinate projections:

$$\varphi_1(x, y, z) = x, \quad \varphi_2(x, y, z) = y, \quad \varphi_3(x, y, z) = z.$$

These are “row vectors” $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$.

(2) The matrix of T is:

$$A = \mathcal{M}(T) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

The matrix of T^* is the transpose:

$$A^T = \mathcal{M}(T^*) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Explicitly, if ψ_1, ψ_2 are the dual basis of $\mathbb{R}^2$ and $\varphi_1, \varphi_2, \varphi_3$ are the dual basis of $\mathbb{R}^3$:

$$T^*(\psi_1) = \varphi_1 + \varphi_2, \quad T^*(\psi_2) = \varphi_2 + \varphi_3.$$

(3) Suppose T is surjective and $T^*(\psi) = 0$, i.e., $\psi \circ T = 0$. For any $w \in W$, since T is surjective, $w = T(v)$ for some $v \in V$. Then:

$$\psi(w) = \psi(T(v)) = (\psi \circ T)(v) = 0.$$

Since this holds for all $w \in W$, we have $\psi = 0$. Thus $\ker(T^*) = \{0\}$, so T^* is injective.

(4) **Well-defined and linear:** For $v \in V$, $\Phi(v) : V^* \rightarrow k$ is defined by $\Phi(v)(\ell) = \ell(v)$. This is linear in ℓ (check!), so $\Phi(v) \in V^{**}$. Moreover, Φ is linear: $\Phi(v_1 + v_2)(\ell) = \ell(v_1 + v_2) = \ell(v_1) + \ell(v_2) = \Phi(v_1)(\ell) + \Phi(v_2)(\ell)$.

Injective: Suppose $\Phi(v) = 0$, meaning $\ell(v) = 0$ for all $\ell \in V^*$. If $v \neq 0$, extend $\{v\}$ to a basis $\{v, v_2, \dots, v_n\}$. The dual basis element φ_1 satisfies $\varphi_1(v) = 1 \neq 0$, contradiction. So $v = 0$.

Surjective: Since $\dim(V) = \dim(V^*) = \dim(V^{**})$, injectivity implies surjectivity.

4.11.6 The Double Dual and Natural Isomorphisms

We’ve seen that $V \cong V^*$ for finite-dimensional V , but this isomorphism required a choice of basis. The double dual $V^{**} = (V^*)^*$ enjoys a more canonical relationship with V .

Definition 4.11.5: Evaluation map

The **evaluation map** is $\text{ev} : V \rightarrow V^{**}$ defined by

$$\text{ev}(v)(\ell) = \ell(v)$$

for $v \in V$ and $\ell \in V^*$. That is, $\text{ev}(v)$ is the linear functional on V^* that “evaluates at v .”

Proposition 4.11.7 Properties of the evaluation map

The evaluation map $\text{ev} : V \rightarrow V^{**}$ satisfies:

1. ev is linear.
2. ev is injective.
3. If V is finite-dimensional, ev is an isomorphism.

Proof: (1) For $v, w \in V$, $\lambda \in k$, and $\ell \in V^*$:

$$\text{ev}(v + \lambda w)(\ell) = \ell(v + \lambda w) = \ell(v) + \lambda \ell(w) = \text{ev}(v)(\ell) + \lambda \text{ev}(w)(\ell).$$

(2) Suppose $\text{ev}(v) = 0$, meaning $\ell(v) = 0$ for all $\ell \in V^*$. If $v \neq 0$, extend $\{v\}$ to a basis and let ℓ be the dual basis element with $\ell(v) = 1$. Contradiction. So $v = 0$.

(3) For finite-dimensional V : $\dim(V^{**}) = \dim(V^*) = \dim(V)$. Since ev is injective, it’s an isomorphism. $\square$

Note:-

The crucial point: ev is natural—it doesn’t depend on any choices. Given bases $\{e_1, \dots, e_n\}$ of V , dual basis $\{e_1^*, \dots, e_n^*\}$ of V^* , and double dual basis $\{e_1^{**}, \dots, e_n^{**}\}$ of V^{**} , one can verify that $\text{ev}(e_i) = e_i^{**}$. The evaluation map sends each basis element to its “double dual counterpart.”

Note:-

For infinite-dimensional V , the situation is strikingly different. If V has a basis $\{e_i\}_{i \in I}$ indexed by an infinite set I , then every element of V is a finite linear combination $\sum_{i \in I} x_i e_i$ (with only finitely many $x_i \neq 0$). However, elements of V^* can be arbitrary: given any function $\alpha : I \rightarrow k$, the map $\ell_\alpha(\sum x_i e_i) = \sum x_i \alpha_i$ is a well-defined linear functional. Thus $V^* \cong k^I = \prod_{i \in I} k$, the full product, which is much larger than the direct sum $V \cong \bigoplus_{i \in I} k$. The evaluation map $\text{ev} : V \rightarrow V^{**}$ is still injective, but far from surjective.

4.11.7 Annihilators Revisited

The annihilator provides a precise duality between subspaces of V and subspaces of V^* .

Definition 4.11.6: Annihilator

For a subspace $U \subseteq V$, the **annihilator** of U is

$$\text{Ann}(U) = U^\circ = \{\ell \in V^* : \ell(u) = 0 \text{ for all } u \in U\}.$$

Theorem 4.11.2 Dimension of the annihilator

If V is finite-dimensional and $U \subseteq V$ is a subspace, then

$$\dim(\text{Ann}(U)) = \dim(V) - \dim(U).$$

Proof: The restriction map $\rho : V^* \rightarrow U^*$ defined by $\rho(\ell) = \ell|_U$ is a linear map with $\ker(\rho) = \text{Ann}(U)$. We claim ρ is surjective: given $\psi \in U^*$, extend a basis of U to a basis of V , define ℓ to agree with ψ on U and be zero on the complementary basis vectors. Then $\rho(\ell) = \psi$.

By rank-nullity:

$$\dim(V^*) = \dim(\text{Ann}(U)) + \dim(U^*).$$

Since $\dim(V^*) = \dim(V)$ and $\dim(U^*) = \dim(U)$, we get $\dim(\text{Ann}(U)) = \dim(V) - \dim(U)$. $\square$

Theorem 4.11.3 Annihilator isomorphism

For a subspace $U \subseteq V$, there is a natural isomorphism

$$\text{Ann}(U) \cong (V/U)^*.$$

Proof: Define $\Phi : \text{Ann}(U) \rightarrow (V/U)^*$ by $\Phi(\ell)(v + U) = \ell(v)$.

Well-defined: If $v + U = v' + U$, then $v - v' \in U$, so $\ell(v) - \ell(v') = \ell(v - v') = 0$ since $\ell \in \text{Ann}(U)$.

Injective: If $\Phi(\ell) = 0$, then $\ell(v) = 0$ for all $v \in V$, so $\ell = 0$.

Surjective: Given $\psi \in (V/U)^*$, define $\ell = \psi \circ q$ where $q : V \rightarrow V/U$ is the quotient map. Then $\ell \in \text{Ann}(U)$ and $\Phi(\ell) = \psi$. $\square$

4.12 Quotient Spaces

Quotient spaces provide a systematic way to “collapse” a subspace to zero, creating a smaller vector space that captures the structure “perpendicular” to the subspace. This construction is fundamental in linear algebra and pervades abstract algebra.

4.12.1 Construction of the Quotient Space

Definition 4.12.1: Cosets and quotient space

Let V be a vector space over k and $U \subseteq V$ a subspace. For $v \in V$, the **coset** of v modulo U is

$$v + U = \{v + u : u \in U\}.$$

The **quotient space** is the set of all cosets:

$$V/U = \{v + U : v \in V\}.$$

Note:-

Two cosets $v + U$ and $w + U$ are equal if and only if $v - w \in U$. Equivalently, the cosets partition V into equivalence classes under the relation $v \sim w \Leftrightarrow v - w \in U$.

Proposition 4.12.1 Vector space structure on the quotient

The quotient V/U becomes a vector space over k under the operations:

1. **Addition:** $(v + U) + (w + U) = (v + w) + U$.
2. **Scalar multiplication:** $\lambda(v + U) = (\lambda v) + U$.

The zero element is $0 + U = U$, and $-(v + U) = (-v) + U$.

Proof: The key is well-definedness: if $v + U = v' + U$ and $w + U = w' + U$, then $v - v' \in U$ and $w - w' \in U$, so $(v + w) - (v' + w') = (v - v') + (w - w') \in U$, meaning $(v + w) + U = (v' + w') + U$.

Similarly for scalar multiplication: if $v + U = v' + U$, then $v - v' \in U$, so $\lambda(v - v') = \lambda v - \lambda v' \in U$, giving $\lambda v + U = \lambda v' + U$.

The vector space axioms follow from those in V . □

4.12.2 The Quotient Map and Exact Sequences

Definition 4.12.2: Quotient map

The **quotient map** (or **canonical projection**) is $q : V \rightarrow V/U$ defined by $q(v) = v + U$.

Proposition 4.12.2 Properties of the quotient map

The quotient map $q : V \rightarrow V/U$ satisfies:

1. q is linear.
2. q is surjective.
3. $\ker(q) = U$.

Proof: (1) $q(v + \lambda w) = (v + \lambda w) + U = (v + U) + \lambda(w + U) = q(v) + \lambda q(w)$.

(2) Every coset $v + U$ is $q(v)$.

(3) $v \in \ker(q) \Leftrightarrow v + U = 0 + U \Leftrightarrow v \in U$. □

Note:-

This gives a **short exact sequence**:

$$0 \longrightarrow U \xrightarrow{\iota} V \xrightarrow{q} V/U \longrightarrow 0$$

where $\iota : U \hookrightarrow V$ is the inclusion. “Exact” means: $\text{Im}(\iota) = \ker(q)$, ι is injective, and q is surjective. Short exact sequences are a central organizing principle in algebra.

Theorem 4.12.1 Dimension of the quotient space

If V is finite-dimensional and $U \subseteq V$ is a subspace, then

$$\dim(V/U) = \dim(V) - \dim(U).$$

Proof: Apply rank-nullity to $q : V \rightarrow V/U$:

$$\dim(V) = \dim(\ker(q)) + \dim(\text{Im}(q)) = \dim(U) + \dim(V/U).$$

□

Example 4.12.1 (Concrete example)

Let $V = \mathbb{R}^3$ and $U = \text{Span}\{(1, 0, 0)\}$, the x -axis. Then $V/U \cong \mathbb{R}^2$:

The coset $(a, b, c) + U$ consists of all points $\{(a + t, b, c) : t \in \mathbb{R}\}$ —a horizontal line through (a, b, c) .

Two points have the same coset iff they have the same y and z coordinates. The map $(a, b, c) + U \mapsto (b, c)$ is an isomorphism $V/U \xrightarrow{\sim} \mathbb{R}^2$.

4.12.3 Universal Property of the Quotient

The quotient space is characterized by a universal property that explains its importance.

Theorem 4.12.2 Universal property of the quotient

Let $\varphi : V \rightarrow W$ be a linear map with $U \subseteq \ker(\varphi)$. Then there exists a unique linear map $\bar{\varphi} : V/U \rightarrow W$ such that $\varphi = \bar{\varphi} \circ q$:

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ q \downarrow & \nearrow \exists! \bar{\varphi} & \\ V/U & & \end{array}$$

Moreover, $\bar{\varphi}$ is defined by $\bar{\varphi}(v + U) = \varphi(v)$.

Proof: **Well-defined:** If $v + U = v' + U$, then $v - v' \in U \subseteq \ker(\varphi)$, so $\varphi(v) = \varphi(v')$.

Linearity: $\bar{\varphi}((v + U) + \lambda(w + U)) = \bar{\varphi}((v + \lambda w) + U) = \varphi(v + \lambda w) = \varphi(v) + \lambda\varphi(w) = \bar{\varphi}(v + U) + \lambda\bar{\varphi}(w + U)$.

Uniqueness: Any $\psi : V/U \rightarrow W$ with $\varphi = \psi \circ q$ must satisfy $\psi(v + U) = \psi(q(v)) = \varphi(v) = \bar{\varphi}(v + U)$. $\square$

Corollary 4.12.1 First Isomorphism Theorem

For any linear map $\varphi : V \rightarrow W$:

$$V/\ker(\varphi) \cong \text{Im}(\varphi).$$

The isomorphism is $\bar{\varphi} : v + \ker(\varphi) \mapsto \varphi(v)$.

Proof: By the universal property, $\bar{\varphi} : V/\ker(\varphi) \rightarrow W$ is well-defined. Its image is exactly $\text{Im}(\varphi)$. Injectivity: if $\bar{\varphi}(v + \ker(\varphi)) = 0$, then $\varphi(v) = 0$, so $v \in \ker(\varphi)$, meaning $v + \ker(\varphi) = 0 + \ker(\varphi)$. $\square$

Note:-

This theorem is remarkably useful. It says that every surjective linear map $\varphi : V \twoheadrightarrow W$ is “really” a quotient map: $V \rightarrow V/\ker(\varphi) \cong W$. Dually, every injective map $\iota : U \hookrightarrow V$ makes U isomorphic to a subspace of V .

4.12.4 Correspondence Between Subspaces

The quotient construction establishes a bijection between certain subspaces.

Theorem 4.12.3 Lattice correspondence

Let $U \subseteq V$ be a subspace. There is a bijection

$$\{\text{subspaces } W \subseteq V \text{ with } U \subseteq W\} \longleftrightarrow \{\text{subspaces of } V/U\}$$

given by:

- Forward: $W \mapsto W/U = \{w + U : w \in W\}$.
- Backward: $\bar{W} \mapsto q^{-1}(\bar{W}) = \{v \in V : v + U \in \bar{W}\}$.

Moreover, this bijection preserves inclusions: $W_1 \subseteq W_2 \Leftrightarrow W_1/U \subseteq W_2/U$.

Proof: We verify the maps are inverses.

Forward then backward: Starting with $W \supseteq U$, we get W/U , then $q^{-1}(W/U) = \{v \in V : v + U \in W/U\}$. But $v + U \in W/U$ iff $v + U = w + U$ for some $w \in W$, iff $v - w \in U \subseteq W$, iff $v \in W$. So $q^{-1}(W/U) = W$.

Backward then forward: Starting with $\bar{W} \subseteq V/U$, let $W = q^{-1}(\bar{W})$. Note $U = q^{-1}(0) \subseteq q^{-1}(\bar{W}) = W$. Then $W/U = q(W) = q(q^{-1}(\bar{W})) = \bar{W}$ (surjectivity of q ensures $q(q^{-1}(S)) = S$ for any $S \subseteq V/U$). $\square$

Note:-

Maximal proper subspaces of V containing U correspond to maximal proper subspaces of V/U —i.e., hyperplanes (codimension-1 subspaces). This perspective becomes useful in algebraic geometry and representation theory.

4.13 Linear Operators and Endomorphisms

When the domain and codomain of a linear map coincide, we gain additional algebraic structure: the ability to compose maps with themselves.

4.13.1 The Ring of Endomorphisms

Definition 4.13.1: Linear operator / Endomorphism

A **linear operator** or **endomorphism** of a vector space V is a linear map $\varphi : V \rightarrow V$. We write

$$\text{End}(V) = \text{Hom}(V, V)$$

for the set of all endomorphisms of V .

Proposition 4.13.1 Ring structure on $\text{End}(V)$

$\text{End}(V)$ is a **ring** (with identity) under:

- **Addition:** $(\varphi + \psi)(v) = \varphi(v) + \psi(v)$.
- **Multiplication:** $(\varphi \circ \psi)(v) = \varphi(\psi(v))$ (composition).
- **Identity:** The identity map id_V .

Moreover, $\text{End}(V)$ is a k -algebra: multiplication distributes over scalar multiplication.

Proof: Addition makes $\text{End}(V)$ an abelian group (it's a subgroup of $\text{Hom}(V, V)$ as a vector space). Composition is associative with identity id_V . Distributivity: $\varphi \circ (\psi + \rho) = \varphi \circ \psi + \varphi \circ \rho$ and $(\varphi + \psi) \circ \rho = \varphi \circ \rho + \psi \circ \rho$. These follow from linearity. $\square$

Note:-

The ring $\text{End}(V)$ is noncommutative unless $\dim(V) \leq 1$. For $V = k^n$, $\text{End}(V) \cong M_n(k)$, the ring of $n \times n$ matrices. The noncommutativity of matrix multiplication reflects the noncommutativity of $\text{End}(V)$.

4.13.2 Polynomials of Operators

Since endomorphisms can be composed, we can form “powers” and evaluate polynomials.

Definition 4.13.2: Powers and polynomials of operators

For $\varphi \in \text{End}(V)$ and $n \in \mathbb{Z}_{\geq 0}$:

$$\varphi^0 = \text{id}_V, \quad \varphi^n = \underbrace{\varphi \circ \varphi \circ \cdots \circ \varphi}_{n \text{ times}}.$$

For a polynomial $p(x) = a_0 + a_1x + \cdots + a_nx^n \in k[x]$, define

$$p(\varphi) = a_0\text{id}_V + a_1\varphi + \cdots + a_n\varphi^n \in \text{End}(V).$$

Proposition 4.13.2 Evaluation homomorphism

The map $\text{ev}_\varphi : k[x] \rightarrow \text{End}(V)$ defined by $p(x) \mapsto p(\varphi)$ is a ring homomorphism. That is:

1. $(p + q)(\varphi) = p(\varphi) + q(\varphi)$.
2. $(pq)(\varphi) = p(\varphi) \circ q(\varphi)$.
3. $1(\varphi) = \text{id}_V$.

Proof: (1) and (3) are immediate. For (2), let $p(x) = \sum a_i x^i$ and $q(x) = \sum b_j x^j$. Then $(pq)(x) = \sum_k c_k x^k$ where $c_k = \sum_{i+j=k} a_i b_j$. So

$$(pq)(\varphi) = \sum_k c_k \varphi^k = \sum_k \left(\sum_{i+j=k} a_i b_j \right) \varphi^k = \left(\sum_i a_i \varphi^i \right) \circ \left(\sum_j b_j \varphi^j \right) = p(\varphi) \circ q(\varphi).$$

□

Note:-

This makes $\text{End}(V)$ a $k[x]$ -module: $p(x) \cdot v = p(\varphi)(v)$. The study of this module structure (particularly the kernel of ev_φ , called the minimal polynomial) is central to the theory of canonical forms.

4.14 Invariant Subspaces

Understanding an operator often begins with finding subspaces on which its behavior is simpler.

Definition 4.14.1: Invariant subspace

A subspace $W \subseteq V$ is **invariant** under $\varphi \in \text{End}(V)$ (or **φ -invariant**) if

$$\varphi(W) \subseteq W,$$

i.e., $\varphi(w) \in W$ for all $w \in W$.

Example 4.14.1 (Basic examples of invariant subspaces)

For any $\varphi \in \text{End}(V)$:

- $\{0\}$ and V are always φ -invariant.
- $\ker(\varphi)$ is φ -invariant: if $\varphi(v) = 0$, then $\varphi(\varphi(v)) = \varphi(0) = 0$.
- $\text{Im}(\varphi)$ is φ -invariant: $\varphi(\text{Im}(\varphi)) = \text{Im}(\varphi^2) \subseteq \text{Im}(\varphi)$.
- More generally, $\ker(\varphi^n)$ and $\text{Im}(\varphi^n)$ are φ -invariant for all $n \geq 0$.

Proposition 4.14.1 Restriction to an invariant subspace

If $W \subseteq V$ is φ -invariant, the **restriction** $\varphi|_W : W \rightarrow W$ is a well-defined linear operator on W .

4.14.1 Block Diagonal Structure

Invariant subspaces lead to simpler matrix representations.

Theorem 4.14.1 Direct sum of invariant subspaces

Suppose $V = V_1 \oplus V_2$ where both V_1 and V_2 are φ -invariant. Choose bases $\mathcal{B}_1 = \{e_1, \dots, e_m\}$ of V_1 and $\mathcal{B}_2 = \{e_{m+1}, \dots, e_n\}$ of V_2 , forming a basis $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2$ of V . Then the matrix of φ with respect to $\mathcal{B}$ is **block diagonal**:

$$\mathcal{M}(\varphi; \mathcal{B}) = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

where $A_1 = \mathcal{M}(\varphi|_{V_1}; \mathcal{B}_1)$ and $A_2 = \mathcal{M}(\varphi|_{V_2}; \mathcal{B}_2)$.

Proof: For $i \leq m$: $\varphi(e_i) \in V_1$ (by φ -invariance), so $\varphi(e_i)$ is a linear combination of $e_1, \dots, e_m$ only. For $i > m$: $\varphi(e_i) \in V_2$, so $\varphi(e_i)$ is a linear combination of $e_{m+1}, \dots, e_n$ only. This gives the block structure. $\square$

Note:-

Conversely, if a matrix is block diagonal with blocks A_1 and A_2 , then the spans of the corresponding basis vectors are invariant subspaces. The goal of much of spectral theory is to decompose V into a direct sum of invariant subspaces on which φ acts simply.

Corollary 4.14.1 Block triangular from one invariant subspace

If V_1 is φ -invariant (but we don't assume V_1 has an invariant complement), then extending a basis of V_1 to a basis of V gives a **block upper-triangular** matrix:

$$\mathcal{M}(\varphi) = \begin{pmatrix} A_1 & * \\ 0 & A_2 \end{pmatrix}.$$

The upper-right block $*$ may be nonzero; it measures the “interaction” between V_1 and V/V_1 .

4.15 Eigenvalues and Eigenvectors

The simplest invariant subspaces are one-dimensional. These give rise to the fundamental notion of eigenvalues.

4.15.1 Definitions and Basic Properties

Definition 4.15.1: Eigenvector and eigenvalue

Let $\varphi \in \text{End}(V)$. A nonzero vector $v \in V$ is an **eigenvector** of φ if

$$\varphi(v) = \lambda v$$

for some scalar $\lambda \in k$. The scalar λ is called the **eigenvalue** corresponding to v .

Note:-

The condition $\varphi(v) = \lambda v$ says that φ acts on the line spanned by v as scaling by λ . Equivalently, $\text{Span}\{v\}$ is a one-dimensional φ -invariant subspace. Conversely, any one-dimensional invariant subspace $U = \text{Span}\{v\}$ consists of eigenvectors (with the same eigenvalue) plus zero.

Proposition 4.15.1 Characterization of eigenvalues

For $\varphi \in \text{End}(V)$ and $\lambda \in k$, the following are equivalent:

1. λ is an eigenvalue of φ .
2. $\varphi - \lambda \text{id}_V$ is not injective.

3. $\ker(\varphi - \lambda \text{id}_V) \neq \{0\}$.
4. (If V is finite-dimensional) $\varphi - \lambda \text{id}_V$ is not surjective.
5. (If V is finite-dimensional) $\det(\varphi - \lambda \text{id}_V) = 0$.

Proof: (1) $\Leftrightarrow$ (2) $\Leftrightarrow$ (3): λ is an eigenvalue iff there exists nonzero v with $\varphi(v) = \lambda v$, iff $(\varphi - \lambda \text{id})(v) = 0$ for some nonzero v , iff $\ker(\varphi - \lambda \text{id}) \neq \{0\}$.

(3) $\Leftrightarrow$ (4) $\Leftrightarrow$ (5): In finite dimensions, a linear operator is injective iff surjective iff invertible iff has nonzero determinant. So $\ker(\varphi - \lambda \text{id}) \neq \{0\}$ iff $\det(\varphi - \lambda \text{id}) = 0$. $\square$

Definition 4.15.2: Eigenspace

For an eigenvalue λ of φ , the **eigenspace** is

$$E(\lambda, \varphi) = \ker(\varphi - \lambda \text{id}_V) = \{v \in V : \varphi(v) = \lambda v\}.$$

This is a subspace of V (it includes 0, though 0 is not an eigenvector).

Theorem 4.15.1 Eigenvectors with distinct eigenvalues are linearly independent

Let $v_1, \dots, v_m$ be eigenvectors of φ with distinct eigenvalues $\lambda_1, \dots, \lambda_m$. Then $v_1, \dots, v_m$ are linearly independent.

Proof: Induction on m . Base case $m = 1$: a single eigenvector is nonzero, hence linearly independent. Suppose the result holds for $m - 1$ eigenvectors. Assume $\sum_{i=1}^m c_i v_i = 0$. Apply φ :

$$\sum_{i=1}^m c_i \lambda_i v_i = 0.$$

Multiply the original equation by λ_m and subtract:

$$\sum_{i=1}^{m-1} c_i (\lambda_i - \lambda_m) v_i = 0.$$

By the induction hypothesis (since $v_1, \dots, v_{m-1}$ are eigenvectors with distinct eigenvalues $\lambda_1, \dots, \lambda_{m-1}$), we have $c_i (\lambda_i - \lambda_m) = 0$ for $i < m$. Since $\lambda_i \neq \lambda_m$, we get $c_i = 0$. Substituting back: $c_m v_m = 0$, so $c_m = 0$ (since $v_m \neq 0$). $\square$

Corollary 4.15.1 Bound on number of eigenvalues

A linear operator on an n -dimensional space has at most n distinct eigenvalues.

Proof: If $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues with eigenvectors $v_1, \dots, v_m$, then $v_1, \dots, v_m$ are linearly independent. In an n -dimensional space, any linearly independent set has at most n elements. $\square$

Corollary 4.15.2 Sum of eigenspaces is direct

If $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues of φ , then

$$E(\lambda_1, \varphi) + \dots + E(\lambda_m, \varphi) = E(\lambda_1, \varphi) \oplus \dots \oplus E(\lambda_m, \varphi).$$

is a direct sum. Furthermore, $\sum_i \dim E(\lambda_i, \varphi) \leq \dim V$.

4.15.2 Diagonalizable Operators

The best-case scenario: V has a basis of eigenvectors.

Definition 4.15.3: Diagonalizable operator

An operator $\varphi \in \text{End}(V)$ is **diagonalizable** if there exists a basis of V consisting entirely of eigenvectors of φ . Equivalently, φ has a diagonal matrix with respect to some basis.

Note:-

If $\{v_1, \dots, v_n\}$ is a basis of eigenvectors with $\varphi(v_i) = \lambda_i v_i$, then the matrix of φ is

$$\mathcal{M}(\varphi) = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

The eigenvalues appear on the diagonal (possibly with repetition).

Theorem 4.15.2 Characterizations of diagonalizability

For $\varphi \in \text{End}(V)$ with $\dim(V) = n$ and distinct eigenvalues $\lambda_1, \dots, \lambda_m$, the following are equivalent:

1. φ is diagonalizable.
2. V has a basis of eigenvectors of φ .
3. $V = E(\lambda_1, \varphi) \oplus \dots \oplus E(\lambda_m, \varphi)$.
4. $\dim E(\lambda_1, \varphi) + \dots + \dim E(\lambda_m, \varphi) = n$.

Proof: (1) $\Leftrightarrow$ (2): By definition.

(2) $\Rightarrow$ (3): If V has a basis of eigenvectors, partition it by eigenvalue. Each part is a basis of the corresponding eigenspace, and together they span V .

(3) $\Rightarrow$ (4): Take dimensions: $\dim V = \dim(E_1 \oplus \dots \oplus E_m) = \sum \dim E_i$.

(4) $\Rightarrow$ (2): The sum $E_1 + \dots + E_m$ is direct (earlier corollary), and has dimension $\sum \dim E_i = n = \dim V$. So $E_1 \oplus \dots \oplus E_m = V$. Combining bases of each E_i gives a basis of V consisting of eigenvectors. $\square$

Corollary 4.15.3 Enough distinct eigenvalues implies diagonalizability

If $\varphi \in \text{End}(V)$ has $\dim(V)$ distinct eigenvalues, then φ is diagonalizable.

Proof: Each eigenspace has dimension at least 1 (it contains an eigenvector). With $n = \dim V$ distinct eigenvalues, $\sum \dim E(\lambda_i) \geq n$. But this sum is also $\leq n$, so equality holds, and criterion (4) above is satisfied. $\square$

4.15.3 Examples: Diagonalizable and Non-Diagonalizable

Example 4.15.1 (Diagonal matrices)

The identity matrix I_n has eigenvalue 1 with eigenspace all of k^n . More generally, a diagonal matrix $\text{diag}(\lambda_1, \dots, \lambda_n)$ has the standard basis vectors as eigenvectors with eigenvalues $\lambda_1, \dots, \lambda_n$. Such a matrix is (trivially) diagonalizable—it's already diagonal!

Example 4.15.2 (A non-diagonalizable matrix)

Consider $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The eigenvalues satisfy $\det(A - \lambda I) = (1 - \lambda)^2 = 0$, so $\lambda = 1$ is the only eigenvalue.

The eigenspace: $E(1, \varphi) = \ker(A - I) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \text{Span}\{\mathbf{e}_1\}$.

This has dimension 1, but $\dim(\mathbb{R}^2) = 2$. So φ is **not diagonalizable**.

Geometrically, φ is a “shear”: it fixes the x -axis pointwise and slides other points parallel to the x -axis.

The only eigenvector direction is along the x -axis.

Example 4.15.3 (No eigenvectors over $\mathbb{R}$)

Consider rotation by 90° in $\mathbb{R}^2$:

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The characteristic polynomial is $\det(R - \lambda I) = \lambda^2 + 1$, which has no real roots. So R has no eigenvalues over $\mathbb{R}$, hence no eigenvectors. Over $\mathbb{C}$, the eigenvalues are $\pm i$ with eigenvectors $(1, \mp i)^T$, and R is diagonalizable as a complex matrix. The eigenfairy doesn’t visit $\mathbb{R}$ (sad).

This illustrates that diagonalizability depends on the scalar field!

Note:-

The rotation example shows that not every operator is diagonalizable. Even over $\mathbb{C}$, where every polynomial has roots, an operator may fail to be diagonalizable (like the shear above). Understanding non-diagonalizable operators requires the theory of Jordan normal form, which we’ll develop later.

Question 5: Exercises on quotient spaces, operators, and eigenvalues

1. Let $V = \mathbb{R}^4$ and $U = \{(x, y, z, w) : x + y = 0, z = w\}$. Find $\dim(U)$, $\dim(V/U)$, and describe V/U explicitly.
2. Prove that if $\varphi, \psi \in \text{End}(V)$ commute ($\varphi \circ \psi = \psi \circ \varphi$), then every eigenspace of φ is ψ -invariant.
3. Let $\varphi \in \text{End}(V)$ be diagonalizable with eigenvalues $\lambda_1, \dots, \lambda_m$. Prove that $p(\varphi) = 0$ where $p(x) = \prod_{i=1}^m (x - \lambda_i)$.
4. Show that the quotient map $q : V \rightarrow V/U$ induces a bijection between $\{\text{linear maps } V/U \rightarrow W\}$ and $\{\text{linear maps } V \rightarrow W \text{ vanishing on } U\}$.
5. Let $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$. Find the eigenvalues and eigenspaces of A . Is A diagonalizable? If so, find a matrix P such that $P^{-1}AP$ is diagonal.

Solution

(1) The constraints $x + y = 0$ and $z = w$ give 2 independent linear equations on $\mathbb{R}^4$, so $\dim(U) = 4 - 2 = 2$. A basis: $\{(-1, 1, 0, 0), (0, 0, 1, 1)\}$.

By the dimension formula: $\dim(V/U) = \dim(V) - \dim(U) = 4 - 2 = 2$.

The map $\pi : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ defined by $\pi(x, y, z, w) = (x + y, z - w)$ has $\ker(\pi) = U$. By the first isomorphism theorem, $V/U \cong \mathbb{R}^2$. Explicitly, the coset $(x, y, z, w) + U$ is determined by $(x + y, z - w)$.

(2) Let v be an eigenvector of φ with eigenvalue λ : $\varphi(v) = \lambda v$. We need $\psi(v) \in E(\lambda, \varphi)$, i.e., $\varphi(\psi(v)) = \lambda \psi(v)$.

Compute: $\varphi(\psi(v)) = \psi(\varphi(v)) = \psi(\lambda v) = \lambda \psi(v)$. ✓

(We used commutativity in the first equality.)

(3) Let $p(x) = \prod_{i=1}^m (x - \lambda_i)$. Since φ is diagonalizable, $V = \bigoplus_{i=1}^m E(\lambda_i, \varphi)$. For any eigenvector $v \in E(\lambda_j, \varphi)$:

$$(\varphi - \lambda_j \text{id})(v) = 0.$$

So $p(\varphi)(v) = (\varphi - \lambda_1) \cdots (\varphi - \lambda_m)(v) = 0$ because one factor gives zero.

Since this holds for all eigenvectors and they span V , we have $p(\varphi) = 0$.

(4) Define $\Phi : \text{Hom}(V/U, W) \rightarrow \{T \in \text{Hom}(V, W) : U \subseteq \ker(T)\}$ by $\Phi(\bar{T}) = \bar{T} \circ q$.

Injective: If $\bar{T} \circ q = 0$, then $\bar{T}(v + U) = \bar{T}(q(v)) = 0$ for all v , so $\bar{T} = 0$.

Surjective: Given $T : V \rightarrow W$ with $U \subseteq \ker(T)$, define $\bar{T} : V/U \rightarrow W$ by $\bar{T}(v + U) = T(v)$. Well-defined because $v - v' \in U \Rightarrow T(v) = T(v')$. Then $\Phi(\bar{T}) = \bar{T} \circ q = T$.

(5) Characteristic polynomial: $\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2)$.

Eigenvalues: $\lambda_1 = 5, \lambda_2 = 2$.

Eigenspaces:

- $\lambda = 5$: $(A - 5I)v = 0$ gives $\begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0$, so $x = 2y$. Eigenspace: $\text{Span}\{(2, 1)^T\}$.

- $\lambda = 2$: $(A - 2I)v = 0$ gives $\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0$, so $x = -y$. Eigenspace: $\text{Span}\{(1, -1)^T\}$.

Since we have 2 linearly independent eigenvectors in $\mathbb{R}^2$, A is diagonalizable.

Let $P = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$ (columns are eigenvectors). Then $P^{-1}AP = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$.

4.16 Let's get general up in here!